Linear Algebra and Its Applications

Chapter 7

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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7.1 Diagonalization of symmetric matrices

A real matrix is symmetric when $A^T = A$, and an orthogonal matrix satisfies $P^T P = I$, so $P^{-1} = P^T$. These are different definitions. The spectral theorem says that a real square matrix has an orthonormal eigenvector basis exactly when it is symmetric. To orthogonally diagonalize a symmetric matrix:

1. Find its real eigenvalues, with multiplicities.
2. For each λ, solve $(A - \lambda I)v = 0$ by row reduction.
3. Within a repeated eigenspace, use Gram–Schmidt if needed. Eigenvectors belonging to distinct eigenvalues are already perpendicular.
4. Normalize each nonzero vector v_i to $u_i = v_i/\|v_i\|$. Set $P = [u_1 \ \cdots \ u_n]$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ in the same order. Then $A = PDP^T$.

In the computations below, the columns of V are unnormalized orthogonal eigenvectors. A diagonal factor rescales the columns to form P. Every displayed diagonalization was checked exactly by $AP = PD$ and $P^T P = I$. Changing a column's sign or reordering columns together with D gives another equally valid answer.

Practice problem 7.1.P1: A square of a symmetric matrix is symmetric

Using the reverse-order transpose rule, $(A^2)^T = (AA)^T = A^T A^T = AA = A^2$, since $A^T = A$. Thus A^2 is symmetric. The reversal creates no difficulty here because the two factors are the same matrix.

Practice problem 7.1.P2: Square an orthogonal diagonalization

If $A = PDP^T$ with $P^T P = I$, then $A^2 = PDP^T PDP^T = PD^2 P^T$. The same P is orthogonal and D^2 is diagonal, so this is an orthogonal diagonalization of A^2.

Exercise 7.1.1: Match entries across the diagonal

The two off-diagonal entries are both 5, so $\begin{bmatrix} 3 & 5 \\ 5 & -7 \end{bmatrix}^T$ is the original matrix. It is symmetric.

Exercise 7.1.2: Opposite signs do not match

The upper-right entry is 5 and the lower-left entry is -5. They would have to be equal for symmetry. Therefore the matrix is not symmetric.

Exercise 7.1.3: One unequal pair is enough

Here $a_{12} = 2$ but $a_{21} = 4$. Thus $A^T \neq A$, so the matrix is not symmetric, despite the repeated entries within each row.

Exercise 7.1.4: Check all three off-diagonal pairs

The paired entries are $a_{12} = a_{21} = 8$, $a_{13} = a_{31} = 3$, and $a_{23} = a_{32} = -2$. The diagonal is unchanged by transpose. Thus this matrix is symmetric.

Exercise 7.1.5: An upper triangular matrix with nonzero off-diagonal entries

The entry $a_{12} = 2$ has counterpart $a_{21} = 0$. Thus the matrix is not symmetric. An upper triangular symmetric matrix would have to be diagonal, because its entries below and above the diagonal must both vanish.

Exercise 7.1.6: Symmetry requires a square matrix

The given array has three rows and four columns. Its transpose has four rows and three columns, so the two cannot be equal. It is not symmetric. The repeated numerical pattern does not overcome the shape mismatch.

Exercise 7.1.7: Orthonormal decimal columns

The two columns have squared lengths $.6^2 + .8^2 = 1$ and $.8^2 + (-.6)^2 = 1$. Their dot product is $.6(.8) + .8(-.6) = 0$. Thus A is orthogonal, and

$$A^{-1} = A^T = \begin{bmatrix} .6 & .8 \\ .8 & -.6 \end{bmatrix}.$$

Here the matrix happens to be symmetric too, so it is its own inverse.

Exercise 7.1.8: A rotation has its transpose as inverse

Each column has squared length $1/2 + 1/2 = 1$, and the dot product is $-1/2 + 1/2 = 0$. Hence the matrix is orthogonal and

$$A^{-1} = A^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Geometrically this reverses the 45° rotation of the original matrix.

Exercise 7.1.9: Perpendicular but not unit columns

The columns $(-5, 2)^T$ and $(2, 5)^T$ are perpendicular, but each has squared length $25 + 4 = 29$. Thus $A^T A = 29I$, not I. The matrix is not orthogonal. Having an inverse is a separate issue; orthogonality specifically requires the columns to be normalized.

Exercise 7.1.10: Another missing normalization

Each column has squared length $1 + 4 + 4 = 9$, and distinct columns have dot product $-2 - 2 + 4 = 0$. Therefore $A^T A = 9I$, not I. The given matrix is not orthogonal; dividing it by 3 would make it so.

Exercise 7.1.11: Verify a less obvious orthogonal matrix

Let the columns be c_1, c_2, c_3. Their squared lengths are

$$\|c_1\|^2 = 4/9 + 5/9 = 1, \quad \|c_2\|^2 = 4/9 + 1/5 + 16/45 = 1, \quad \|c_3\|^2 = 1/9 + 4/5 + 4/45 = 1.$$

Their pairwise dot products are $4/9 - 4/9 = 0$, $2/9 - 2/9 = 0$, and $2/9 - 2/5 + 8/45 = 0$. Thus all columns are orthonormal and the inverse is the transpose:

$$A^{-1} = \begin{bmatrix} 2/3 & 0 & \sqrt{5}/3 \\ 2/3 & 1/\sqrt{5} & -4/\sqrt{45} \\ 1/3 & -2/\sqrt{5} & -2/\sqrt{45} \end{bmatrix}.$$

Exercise 7.1.12: An orthogonal signed matrix

Write $A = H/2$, where H has the printed entries 1 or -1. Each column of H has squared length 4, and every pair of different columns has two agreeing signs and two disagreeing signs, so dot product zero. Thus $A^T A = I$, and

$$A^{-1} = A^T = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}.$$

Exercise 7.1.13: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t-4)(t-2)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = 2$: row reduction gives $x_1 = -x_2$. $\lambda = 4$: row reduction gives $x_1 = x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (2, 4).$$

Their squared lengths are $(2, 2)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{2}} V, \qquad D = \operatorname{diag}(2, 4).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.14: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t - 6)(t + 4)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = -4$: row reduction gives $x_1 = -x_2$. $\lambda = 6$: row reduction gives $x_1 = x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (-4, 6).$$

Their squared lengths are $(2, 2)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{2}} V, \qquad D = \operatorname{diag}(-4, 6).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.15: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $t(t - 17)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = 0$: row reduction gives $x_1 = \frac{x_2}{4}$. $\lambda = 17$: row reduction gives $x_1 = -4x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 4 \\ 4 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (0, 17).$$

Their squared lengths are $(17, 17)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{17}} V, \qquad D = \operatorname{diag}(0, 17).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.16: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t - 25)(t + 25)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = -25$: row reduction gives $x_1 = -\frac{4x_2}{3}$. $\lambda = 25$: row reduction gives $x_1 = \frac{3x_2}{4}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (-25, 25).$$

Their squared lengths are $(25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(-25, 25).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.17: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -2$: row reduction gives $x_1 = -x_3, \quad x_2 = 0$. $\lambda = 2$: row reduction gives $x_1 = x_3, \quad x_2 = -2x_3$. $\lambda = 5$: row reduction gives $x_1 = x_3, \quad x_2 = x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-2, 2, 5).$$

Their squared lengths are $(2, 6, 3)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{3}), \qquad D = \operatorname{diag}(-2, 2, 5).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.18: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -50$: row reduction gives $x_1 = \frac{3x_2}{4}, \quad x_3 = 0$. $\lambda = 3$: row reduction gives $x_1 = 0, \quad x_2 = 0$. $\lambda = 25$: row reduction gives $x_1 = -\frac{4x_2}{3}, \quad x_3 = 0$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 3 & 0 & 4 \\ 4 & 0 & -3 \\ 0 & 1 & 0 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-50, 3, 25).$$

Their squared lengths are $(25, 1, 25)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{1}{5}, 1, \frac{1}{5}), \qquad D = \operatorname{diag}(-50, 3, 25).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.19: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -2$: row reduction gives $x_1 = -x_3, \quad x_2 = -\frac{x_3}{2}$. $\lambda = 7$: row reduction gives $x_1 = -\frac{x_2}{2} + x_3$. Within the 7-eigenspace, start with $(1, 0, 1)^T$ and $(-1/2, 1, 0)^T$. Subtracting the projection of the latter onto the former gives $(-1/4, 1, 1/4)^T$, a multiple of $(-1, 4, 1)^T$. This is the Gram–Schmidt step that produces the two orthogonal columns used below.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 4 \\ -2 & 1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-2, 7, 7).$$

Their squared lengths are $(9, 2, 18)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{1}{3}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{6}), \qquad D = \operatorname{diag}(-2, 7, 7).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.20: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = -2x_3, \quad x_2 = -2x_3$. $\lambda = 7$: row reduction gives $x_1 = -\frac{x_3}{2}, \quad x_2 = x_3$. $\lambda = 13$: row reduction gives $x_1 = x_3, \quad x_2 = -\frac{x_3}{2}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 1 & 2 \\ 2 & -2 & -1 \\ -1 & -2 & 2 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (1, 7, 13).$$

Their squared lengths are $(9, 9, 9)$. Hence an explicit answer is

$$P = \frac{1}{3}V, \qquad D = \operatorname{diag}(1, 7, 13).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.21: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = -x_3, \quad x_2 = -x_4$. $\lambda = 5$: row reduction gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = -x_4$. $\lambda = 9$: row reduction gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ -1 & 0 & 1 & 1 \\ 0 & -1 & -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (1, 1, 5, 9).$$

Their squared lengths are $(2, 2, 4, 4)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, \frac{1}{2}), \qquad D = \operatorname{diag}(1, 1, 5, 9).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.22: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 0$: row reduction gives $x_1 = 0, \quad x_2 = -x_4, \quad x_3 = 0$. $\lambda = 2$: row reduction gives $x_2 = x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (0, 2, 2, 2).$$

Their squared lengths are $(2, 1, 1, 2)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, 1, 1, \frac{\sqrt{2}}{2}), \qquad D = \operatorname{diag}(0, 2, 2, 2).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.23: An orthogonal eigenvector basis

The printed vector $v = (1, 1, 1)^T$ satisfies $Av = 5v$, so its eigenvalue is 5. The exercise also asks to verify 2 is an eigenvalue; the vector $(1, -1, 0)^T$ satisfies $A(1, -1, 0)^T = 2(1, -1, 0)^T$. Thus the two assertions in the wording refer to different eigenvectors.

Solve the eigenspace equations.

$\lambda = 2$: row reduction gives $x_1 = -x_2 - x_3$. $\lambda = 5$: row reduction gives $x_1 = x_3, \quad x_2 = x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (2, 2, 5).$$

Their squared lengths are $(2, 6, 3)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{3}), \qquad D = \operatorname{diag}(2, 2, 5).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.24: An orthogonal eigenvector basis

Multiplication gives $Av_1 = 10v_1$ for the supplied $v_1 = (-2, 2, 1)^T$, and $Av_2 = v_2$ for $v_2 = (1, 1, 0)^T$. For the repeated eigenvalue 1, $(1, -1, 4)^T$ is a second eigenvector orthogonal to v_2.

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = x_2 + \frac{x_3}{2}$. $\lambda = 10$: row reduction gives $x_1 = -2x_3, \quad x_2 = 2x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & -2 \\ 0 & 4 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (1, 1, 10).$$

Their squared lengths are $(2, 18, 9)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{6}, \frac{1}{3}), \qquad D = \operatorname{diag}(1, 1, 10).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.25: Symmetry, multiplicity, and a missing unit condition

(a) **True.** If $A = PDP^T$, then $A^T = (PDP^T)^T = PD^T P^T = PDP^T = A$.

(b) **True.** Symmetry gives $3u^T v = (Au)^T v = u^T Av = 4u^T v$, so $u^T v = 0$.

(c) **False.** I_n is symmetric but has only one distinct eigenvalue, namely 1 with multiplicity n.

(d) **False.** For $B = vv^T$, $B^2 = (v^T v)B$. A projection requires $B^2 = B$, which for nonzero v requires $\|v\| = 1$. For $v = 2e_1$, $B = 4e_1 e_1^T$ is not idempotent.

Exercise 7.1.26: Four consequences and nonconsequences of the spectral theorem

(a) **True.** This is the real symmetric spectral theorem.

(b) **True.** Taking the transpose of PDP^T gives the same matrix because D is real diagonal.

(c) **False.** A 90° rotation $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is orthogonal but not symmetric, so it cannot be orthogonally diagonalized over $\mathbb{R}$.

(d) **True.** In the spectral theorem the orthogonal diagonalization provides one basis vector for each occurrence of an eigenvalue on the diagonal, giving equality of geometric and algebraic multiplicities.

Exercise 7.1.27: Three symmetric products

Use the reverse-order transpose rule and the hypothesis $A^T = A$:

$$(B^T AB)^T = B^T A^T B = B^T AB, \quad (B^T B)^T = B^T B, \quad (BB^T)^T = BB^T.$$

The products have square sizes $m \times m$, $m \times m$, and $n \times n$, respectively, so these equalities indeed establish symmetry in each case.

Exercise 7.1.28: Move a symmetric matrix across a dot product

For every x, y, $(Ax) \cdot y = (Ax)^T y = x^T A^T y = x^T Ay = x \cdot (Ay)$. The step using $A^T = A$ is exactly where symmetry matters. Without it the correct expression would contain A^T on the other side.

Exercise 7.1.29: The inverse retains the eigenvectors

Write $A = PDP^T$ with P orthogonal. Since A is invertible, none of the diagonal eigenvalues in D is zero, so D^{-1} exists and is diagonal. Then $A^{-1} = PD^{-1}P^T$, an orthogonal diagonalization with the same P. Its diagonal eigenvalues are the reciprocals of the original ones.

Exercise 7.1.30: A commuting product remains symmetric

Orthogonal diagonalizability makes both A and B symmetric. Therefore $(AB)^T = B^T A^T = BA = AB$, using the additional commutation hypothesis. Thus AB is symmetric and the spectral theorem makes it orthogonally diagonalizable. Without $AB = BA$, a product of two symmetric matrices need not be symmetric.

Exercise 7.1.31: Count a repeated eigenvalue's eigenspace dimension

Let $x = Py$. Since P is invertible, $Ax = \lambda x$ is equivalent to $(D - \lambda I)y = 0$. At positions where the diagonal of D differs from λ, this forces $y_i = 0$. At its k positions equal to λ, the corresponding y_i are arbitrary. Thus the solution space has exactly k independent coordinates. Mapping by P gives the eigenspace of A and preserves its dimension, so that dimension is k. Equivalently the k corresponding columns of P form its basis.

Exercise 7.1.32: A symmetric triangular matrix is diagonal

From $A = PRP^T$, solve for $R = P^T AP$. Taking transposes gives $R^T = P^T A^T P = P^T AP = R$, so R is symmetric. But R is also upper triangular. Entries below its diagonal are zero; symmetry forces their counterpart entries above the diagonal to be zero too. Thus all off-diagonal entries vanish and R is diagonal.

Exercise 7.1.33: A spectral decomposition for Example 2

Use the orthogonal eigenvectors $a = (-1, 1, 0)^T$, $b = (-1, -1, 2)^T$, $c = (1, 1, 1)^T$, with eigenvalues $8, 6, 3$ and squared lengths $2, 6, 3$. After normalization the spectral formula gives

$$\boxed{A = 4aa^T + bb^T + cc^T}.$$

Indeed the normalization divides each outer product by its squared length; thus the three coefficients are 8/2, 6/6, 3/3. Expanding these outer products gives $\begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$, the matrix from Example 2. This checks the decomposition entrywise.

Exercise 7.1.34: A spectral decomposition with a repeated eigenvalue

For Example 3, take $a = (1, 0, 1)^T$, $b = (-1, 4, 1)^T$, and $c = (-2, -1, 2)^T$. They are mutually perpendicular, with squared lengths $2, 18, 9$. The first two have eigenvalue 7, and the last has eigenvalue -2. Therefore

$$\boxed{A = \frac{7}{2}aa^T + \frac{7}{18}bb^T - \frac{2}{9}cc^T}.$$

For instance the $(1, 1)$ entry is $7/2 + 7/18 - 8/9 = 3$; expansion of all entries returns $\begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$. The two separate rank-one terms for eigenvalue 7 reflect its two-dimensional eigenspace.

Exercise 7.1.35: A rank-one projection from a unit vector

(a) $Bx = (uu^T)x = u(u^T x) = (x \cdot u)u$. Since $u \cdot u = 1$, this equals $((x \cdot u)/(u \cdot u))u$, the orthogonal projection formula.

(b) $B^T = (uu^T)^T = uu^T = B$ and $B^2 = u(u^T u)u^T = uu^T = B$.

(c) $Bu = u(u^T u) = u$. The unit vector is nonzero, so it is an eigenvector with eigenvalue 1.

Exercise 7.1.36: Every symmetric idempotent matrix is an orthogonal projection

(a) Show the proposed pieces are perpendicular.

Put $\widehat{y} = By$, $z = (I - B)y$. Symmetry and idempotence give

$$z^T \widehat{y} = y^T (I - B)^T By = y^T (B - B^2)y = 0.$$

(b) Verify the stronger orthogonality needed for a projection.

Clearly $\widehat{y} \in W = \operatorname{Col} B$. For any $w \in W$, write $w = Bv$; then $z^T w = y^T (I - B)Bv = 0$. Thus $z \in W^{\perp}$, not merely perpendicular to one particular vector $\widehat{y}$. The identity $y = By + (y - By)$ is therefore the orthogonal decomposition into a vector in W and one in $W^{\perp}$. Its uniqueness proves that $By = \operatorname{proj}_W y$.

Exercise 7.1.37: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -12$: row reduction gives $x_1 = x_4, \quad x_2 = -x_4, \quad x_3 = -x_4$. $\lambda = 4$: row reduction gives $x_1 = -x_4, \quad x_2 = -x_4, \quad x_3 = x_4$. $\lambda = 10$: row reduction gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = 18$: row reduction gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = -x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (-12, 4, 10, 18).$$

Their squared lengths are $(4, 4, 4, 4)$. Hence an explicit answer is

$$P = \frac{1}{2}V, \qquad D = \operatorname{diag}(-12, 4, 10, 18).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.38: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = \frac{1}{4}$: row reduction gives $x_1 = 4x_4, \quad x_2 = 2x_4, \quad x_3 = 2x_4$. $\lambda = \frac{3}{10}$: row reduction gives $x_1 = -\frac{x_4}{4}, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = \frac{x_4}{2}$. $\lambda = \frac{11}{20}$: row reduction gives $x_1 = x_4, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = -2x_4$. $\lambda = \frac{3}{4}$: row reduction gives $x_1 = -x_4, \quad x_2 = 2x_4, \quad x_3 = -\frac{x_4}{2}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 4 & 1 & 2 & 2 \\ 2 & 2 & -1 & -4 \\ 2 & -2 & -4 & 1 \\ 1 & -4 & 2 & -2 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (\frac{1}{4}, \frac{3}{10}, \frac{11}{20}, \frac{3}{4}).$$

Their squared lengths are $(25, 25, 25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(\frac{1}{4}, \frac{3}{10}, \frac{11}{20}, \frac{3}{4}).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.39: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -\frac{5}{4}$: row reduction gives $x_1 = -x_4, \quad x_2 = 2x_4, \quad x_3 = -\frac{x_4}{2}$. $\lambda = 0$: row reduction gives $x_1 = -x_4, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = 2x_4$. $\lambda = \frac{3}{4}$: row reduction gives $x_1 = \frac{3x_3}{2} + x_4, \quad x_2 = x_3$. For the repeated eigenvalue $3/4$, the two chosen null vectors $(4, 2, 2, 1)^T$ and $(1, -2, -2, 4)^T$ have dot product $4 - 4 - 4 + 4 = 0$. Both have squared length 25, so dividing each by 5 normalizes them.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 2 & 4 & 1 \\ -4 & 1 & 2 & -2 \\ 1 & -4 & 2 & -2 \\ -2 & -2 & 1 & 4 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (-\frac{5}{4}, 0, \frac{3}{4}, \frac{3}{4}).$$

Their squared lengths are $(25, 25, 25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(-\frac{5}{4}, 0, \frac{3}{4}, \frac{3}{4}).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.40: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -28$: row reduction gives $x_1 = -\frac{x_5}{4}, \quad x_2 = -\frac{x_5}{4}, \quad x_3 = -\frac{x_5}{4}, \quad x_4 = -\frac{x_5}{4}$. $\lambda = 8$: row reduction gives $x_1 = -x_2 - x_3, \quad x_4 = 0, \quad x_5 = 0$. $\lambda = 17$: row reduction gives $x_1 = x_5, \quad x_2 = x_5, \quad x_3 = x_5, \quad x_4 = x_5$. $\lambda = 32$: row reduction gives $x_1 = -\frac{x_4}{3}, \quad x_2 = -\frac{x_4}{3}, \quad x_3 = -\frac{x_4}{3}, \quad x_5 = 0$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 1 \\ 1 & 0 & -2 & 1 & 1 \\ 1 & 0 & 0 & 1 & -3 \\ -4 & 0 & 0 & 1 & 0 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_5) = (-28, 8, 8, 17, 32).$$

Their squared lengths are $(20, 2, 6, 5, 12)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{5}}{10}, \frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{5}}{5}, \frac{\sqrt{3}}{6}), \qquad D = \operatorname{diag}(-28, 8, 8, 17, 32).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

7.2 Quadratic forms

For a symmetric matrix A, the expression $Q(x) = x^T Ax$ is a quadratic form. Diagonal entries are the coefficients of squares. Off-diagonal entries contribute twice: the coefficient of $x_i x_j$ for $i \neq j$ is $2a_{ij}$. Thus put *half* of each mixed-term coefficient in each of the two symmetric positions.

An orthogonal change $x = Py$ gives $Q(x) = y^T(P^T AP)y$. If P consists of orthonormal eigenvectors, then $P^T AP = D$ is diagonal and $Q = \sum_i \lambda_i y_i^2$. All eigenvalues positive means positive definite; all negative means negative definite. Nonnegative eigenvalues with a zero give positive semidefinite but not definite, and the analogous nonpositive case is negative semidefinite. Both signs give an indefinite form. A definite inequality is required only for $x \neq 0$, since every quadratic form has $Q(0) = 0$.

Practice problem 7.2.P1: Recognize positive semidefiniteness spectrally

A symmetric matrix is positive semidefinite exactly when all eigenvalues are nonnegative. In an orthonormal eigenbasis, $Q(x) = \sum_i \lambda_i y_i^2 \geq 0$ if every $\lambda_i \geq 0$. Conversely, if $\lambda_j < 0$, taking $y = e_j$ and $x = Pe_j$ gives $Q(x) = \lambda_j < 0$, contradicting semidefiniteness. This argument also includes zero eigenvalues and the zero matrix.

Exercise 7.2.1: Evaluate a two-variable form

(a) First multiply $Ax = (5x_1 + x_2/3, x_1/3 + x_2)^T$. Dotting with x gives $Q(x) = 5x_1^2 + (2/3)x_1x_2 + x_2^2$.

(b) For $x = (6, 1)^T$, the value is $5(36) + (2/3)6 + 1 = 185$.

(c) For $x = (1, 3)^T$, the value is $5 + (2/3)3 + 9 = 16$.

The two off-diagonal contributions both matter; omitting one changes the answer.

Exercise 7.2.2: Evaluate a three-variable form

(a) Expanding $x^T Ax$ and collecting the symmetric terms gives $Q(x) = 4x_1^2 + 2x_2^2 + x_3^2 + 6x_1x_2 + 2x_2x_3$. There is no x_1x_3 term because $a_{13} = a_{31} = 0$.

(b) At $x = (2, -1, 5)^T$, this is $16 + 2 + 25 - 12 - 10 = 21$.

(c) At $x = (1, 1, 1)^T/\sqrt{3}$, each square and each pair product is $1/3$. Thus $Q(x) = (4+2+1+6+2)/3 = 5$.

Exercise 7.2.3: Recover two symmetric coefficient matrices

The diagonal entries come from the square terms and each mixed coefficient is divided by two. Thus the matrices are

$$\text{(a)} \quad \boxed{\begin{bmatrix} 10 & -3 \\ -3 & -3 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 5 & 3/2 \\ 3/2 & 0 \end{bmatrix}}.$$

For example the first mixed coefficient is $-3 - 3 = -6$, and the second form has no x_2^2 term, so its bottom-right entry is zero.

Exercise 7.2.4: A large mixed coefficient and a pure mixed term

The same rule gives

$$\text{(a)} \quad \boxed{\begin{bmatrix} 20 & 15/2 \\ 15/2 & -10 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}}.$$

The two $1/2$ entries in part (b) sum to the required coefficient 1 of x_1x_2; neither diagonal entry is needed.

Exercise 7.2.5: Three-variable coefficient matrices

Record the three square coefficients on the diagonal, then place half of each of the three possible mixed coefficients in symmetric positions:

$$\text{(a)} \quad \boxed{\begin{bmatrix} 8 & -3 & 2 \\ -3 & 7 & -1 \\ 2 & -1 & -3 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & 2 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}}.$$

In part (b) all square coefficients vanish, explaining the zero diagonal. Expanding either matrix as x^TAx reproduces every original coefficient.

Exercise 7.2.6: Mixed terms with odd coefficients

Fractions are necessary when a mixed coefficient is odd. The two matrices are

$$\text{(a)} \quad \boxed{\begin{bmatrix} 5 & 5/2 & -3/2 \\ 5/2 & -1 & 0 \\ -3/2 & 0 & 7 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}}.$$

For instance $2(5/2) = 5$ and $2(-3/2) = -3$ give the two mixed coefficients in part (a); the absent x_2x_3 term gives a zero entry.

Exercise 7.2.7: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-4, 6)$ (counted with multiplicity). $\lambda = -4$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_2$. $\lambda = 6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 2)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{2}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-4, 6)$, so

$$Q(x) = \boxed{-4y_1^2 + 6y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.8: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 9 & -4 & 4 \\ -4 & 7 & 0 \\ 4 & 0 & 11 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(3, 9, 15)$ (counted with multiplicity). $\lambda = 3$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_3, \quad x_2 = -2x_3$. $\lambda = 9$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_3}{2}, \quad x_2 = x_3$. $\lambda = 15$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_3, \quad x_2 = -\frac{x_3}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 & 2 \\ 2 & -2 & -1 \\ -1 & -2 & 2 \end{bmatrix}.$$

Their squared lengths are $(9, 9, 9)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{3}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(3, 9, 15)$, so

$$Q(x) = \boxed{3y_1^2 + 9y_2^2 + 15y_3^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.9: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(2, 7)$ (counted with multiplicity). $\lambda = 2$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 2x_2$. $\lambda = 7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(2, 7)$, so

$$Q(x) = \boxed{2y_1^2 + 7y_2^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.10: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 9 & -4 \\ -4 & 3 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(1, 11)$ (counted with multiplicity). $\lambda = 1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{x_2}{2}$. $\lambda = 11$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(1, 11)$, so

$$Q(x) = \boxed{y_1^2 + 11y_2^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.11: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 2 & 5 \\ 5 & 2 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-3, 7)$ (counted with multiplicity). $\lambda = -3$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_2$. $\lambda = 7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 2)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{2}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(-3, 7)$, so

$$Q(x) = \boxed{-3y_1^2 + 7y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.12: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-6, -1)$ (counted with multiplicity). $\lambda = -6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_2$. $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{x_2}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-6, -1)$, so

$$Q(x) = \boxed{-6y_1^2 - y_2^2}.$$

The form is **negative definite**. Every nonzero y has a nonzero squared coordinate with a negative coefficient, so $Q < 0$. Invertibility of P transfers this to nonzero x.

Exercise 7.2.13: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(0, 10)$ (counted with multiplicity). $\lambda = 0$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_2$. $\lambda = 10$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{3}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix}.$$

Their squared lengths are $(10, 10)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{10}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(0, 10)$, so

$$Q(x) = \boxed{10y_2^2}.$$

The form is **positive semidefinite but not positive definite**. All terms are nonnegative, but a nonzero vector along a zero-eigenvalue coordinate gives $Q = 0$, preventing strict positive definiteness.

Exercise 7.2.14: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 8 & 3 \\ 3 & 0 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-1, 9)$ (counted with multiplicity). $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{3}$. $\lambda = 9$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 3 \\ -3 & 1 \end{bmatrix}.$$

Their squared lengths are $(10, 10)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{10}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(-1, 9)$, so

$$Q(x) = \boxed{-y_1^2 + 9y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.15: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} -2 & 2 & 2 & 2 \\ 2 & -6 & 0 & 0 \\ 2 & 0 & -9 & 3 \\ 2 & 0 & 3 & -9 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-12, -8, -6, 0)$ (counted with multiplicity). $\lambda = -12$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = 0, \quad x_3 = -x_4$. $\lambda = -8$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = -6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = -2x_4, \quad x_3 = x_4$. $\lambda = 0$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_4, \quad x_2 = x_4, \quad x_3 = x_4$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & -1 & 2 & 1 \\ 1 & -1 & -1 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 4, 6, 12)$ and their pairwise dot products vanish. Set

$$P = V \operatorname{diag}\left(\frac{\sqrt{2}}{2}, \frac{1}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{6}\right); \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-12, -8, -6, 0)$, so

$$Q(x) = \boxed{-12y_1^2 - 8y_2^2 - 6y_3^2}.$$

The form is **negative semidefinite but not negative definite**. All terms are nonpositive, but a nonzero vector along the zero-eigenvalue coordinate gives $Q = 0$, preventing strict negative definiteness.

Exercise 7.2.16: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 4 & \frac{3}{2} & 0 & -2 \\ \frac{3}{2} & 4 & 2 & 0 \\ 0 & 2 & 4 & \frac{3}{2} \\ -2 & 0 & \frac{3}{2} & 4 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(\frac{3}{2}, \frac{3}{2}, \frac{13}{2}, \frac{13}{2})$ (counted with multiplicity). $\lambda = \frac{3}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} + \frac{5x_4}{4}$, $\quad x_2 = -\frac{5x_3}{4} - \frac{3x_4}{4}$. $\lambda = \frac{13}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} - \frac{5x_4}{4}$, $\quad x_2 = \frac{5x_3}{4} - \frac{3x_4}{4}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 4 & 3 & 4 \\ -5 & 0 & 5 & 0 \\ 4 & -3 & 4 & -3 \\ 0 & 5 & 0 & -5 \end{bmatrix}.$$

Their squared lengths are $(50, 50, 50, 50)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{5\sqrt{2}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(\frac{3}{2}, \frac{3}{2}, \frac{13}{2}, \frac{13}{2})$, so

$$Q(x) = \boxed{\frac{3y_1^2}{2} + \frac{3y_2^2}{2} + \frac{13y_3^2}{2} + \frac{13y_4^2}{2}}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.17: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & \frac{9}{2} & 0 & -6 \\ \frac{9}{2} & 1 & 6 & 0 \\ 0 & 6 & 1 & \frac{9}{2} \\ -6 & 0 & \frac{9}{2} & 1 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-\frac{13}{2}, -\frac{13}{2}, \frac{17}{2}, \frac{17}{2})$ (counted with multiplicity). $\lambda = -\frac{13}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} + \frac{5x_4}{4}, \quad x_2 = -\frac{5x_3}{4} - \frac{3x_4}{4}$. $\lambda = \frac{17}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} - \frac{5x_4}{4}, \quad x_2 = \frac{5x_3}{4} - \frac{3x_4}{4}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 4 & 3 & 4 \\ -5 & 0 & 5 & 0 \\ 4 & -3 & 4 & -3 \\ 0 & 5 & 0 & -5 \end{bmatrix}.$$

Their squared lengths are $(50, 50, 50, 50)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{5\sqrt{2}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T A P = \operatorname{diag}(-\frac{13}{2}, -\frac{13}{2}, \frac{17}{2}, \frac{17}{2})$, so

$$Q(x) = \boxed{-\frac{13y_1^2}{2} - \frac{13y_2^2}{2} + \frac{17y_3^2}{2} + \frac{17y_4^2}{2}}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.18: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 11 & -6 & -6 & -6 \\ -6 & -1 & 0 & 0 \\ -6 & 0 & 0 & -1 \\ -6 & 0 & -1 & 0 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-7, -1, 1, 17)$ (counted with multiplicity). $\lambda = -7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = -2x_4, \quad x_3 = x_4$. $\lambda = 1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = 0, \quad x_3 = -x_4$. $\lambda = 17$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -3x_4, \quad x_2 = x_4, \quad x_3 = x_4$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & 2 & 0 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 \end{bmatrix}.$$

Their squared lengths are $(4, 6, 2, 12)$ and their pairwise dot products vanish. Set

$$P = V \operatorname{diag}(\frac{1}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{6}); \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-7,-1,1,17)$, so

$$Q(x) = \boxed{-7y_1^2 - y_2^2 + y_3^2 + 17y_4^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.19: A unit-circle maximum

Since $x_1^2 + x_2^2 = 1$, $5x_1^2 + 8x_2^2 = 5 + 3x_2^2 \le 8$. Equality occurs at $x = (0,1)$ or $(0,-1)$. Thus the largest value is $\boxed{8}$. The calculation provides both an upper bound and a point attaining it.

Exercise 7.2.20: A negative coefficient lowers the maximum

Again use the constraint to write $5x_1^2 - 3x_2^2 = 5 - 8x_2^2 \le 5$. Equality occurs at $x = (1,0)$ or $(-1,0)$, so the largest value is $\boxed{5}$. Without the unit constraint, scaling x_1 would make the form arbitrarily large.

Exercise 7.2.21: Six true–false statements about quadratic forms

(a) **True.** The matrix of a form in this section means its canonical symmetric coefficient matrix, with mixed coefficients split equally. A nonsymmetric representing array can be replaced by its symmetric part.

(b) **True.** In a symmetric matrix the mixed coefficient is $2a_{ij}$, so all mixed coefficients vanish exactly when every off-diagonal entry vanishes.

(c) **True.** The principal-axis directions are given by an orthonormal eigenvector basis of the symmetric coefficient matrix.

(d) **False.** The condition is $Q(x) > 0$ for all $x \neq 0$. At $x = 0$ every quadratic form equals zero.

(e) **True.** In eigenvector coordinates the form is a sum of positive coefficients times squares, strictly positive for nonzero inputs.

(f) **True.** This is the stated form of a Cholesky factorization. Its existence is restricted to positive definite symmetric matrices; the statement describes the factorization, not a claim that every symmetric matrix admits one.

Exercise 7.2.22: Five further classification statements

(a) **True.** $\|x\|^2 = x^TIx$ is a quadratic form.

(b) **False.** Orthogonality of P alone is insufficient; its columns must be appropriate eigenvectors. For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $P = I$, the mixed term $2y_1y_2$ remains.

(c) **False.** Degenerate or empty level sets are also possible. For $A = I$, $c = -1$ the set is empty; for $A = \operatorname{diag}(1,0)$, $c = 0$ the level set is the line $x_1 = 0$, not one of the three listed curves.

(d) **False.** An indefinite form takes both positive and negative values. Semidefinite forms have only one weak sign. For example $x_1^2 - x_2^2$ is indefinite and neither kind of semidefinite.

(e) **True.** For each unit eigenvector u_i, $u_i^T A u_i = \lambda_i$. Strict negativity at every nonzero input therefore makes every eigenvalue negative.

Exercise 7.2.23: Compare characteristic-polynomial coefficients

For $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$,

$$\det(A - tI) = (a - t)(d - t) - b^2 = t^2 - (a + d)t + (ad - b^2).$$

On the other hand it is $(t - \lambda_1)(t - \lambda_2) = t^2 - (\lambda_1 + \lambda_2)t + \lambda_1\lambda_2$. Equality of polynomial coefficients gives

$$\boxed{\lambda_1 + \lambda_2 = a + d, \qquad \lambda_1\lambda_2 = \det A}.$$

The factor signs agree because there are two factors; in odd dimensions one must distinguish $\det(A - tI)$ from $\det(tI - A)$.

Exercise 7.2.24: Classify nonsingular two-variable forms without solving for roots

(a) If $\det A = ad - b^2 > 0$ and $a > 0$, then $ad > b^2 \geq 0$, so $d > 0$ and $a + d > 0$. The two real eigenvalues have positive product, so they have the same nonzero sign; their positive sum forces both positive. Thus the form is positive definite.

(b) If $\det A > 0$ and $a < 0$, then $ad > 0$ makes $d < 0$, so the eigenvalues have the same nonzero sign and negative sum. Both are negative, giving a negative definite form.

(c) If $\det A < 0$, the real eigenvalues have negative product, so one is positive and one negative. Their eigenvector directions give opposite values of the form, proving indefiniteness.

The hypothesis $\det A \neq 0$ excludes the semidefinite boundary cases.

Exercise 7.2.25: Gram matrices are positive semidefinite

The matrix $B^T B$ is symmetric. For any x, $x^T B^T B x = (Bx)^T(Bx) = \|Bx\|^2 \geq 0$, proving positive semidefiniteness. If B is square and invertible, $x \neq 0$ implies $Bx \neq 0$, making the last value strictly positive. Thus $B^T B$ is positive definite then. In fact independent columns suffice even for a rectangular B.

Exercise 7.2.26: Construct a positive definite square root

Write $A = PDP^T$ with orthogonal P and all $\lambda_i > 0$ on the diagonal of D. Let $C = \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})$, so $C^T C = D$, and set $B = PCP^T$. This B is symmetric and, for nonzero x with $y = P^T x \neq 0$, $x^T B x = \sum_i \sqrt{\lambda_i} y_i^2 > 0$, so B is positive definite. Finally

$$B^T B = PC^T P^T PCP^T = PC^T CP^T = PDP^T = A.$$

Both the factorization and the positive definiteness requested of B have now been verified.

Exercise 7.2.27: A sum of positive definite matrices

The sum $A+B$ is symmetric. For $x \neq 0$, $x^T(A+B)x = x^TAx + x^TBx > 0+0 = 0$, since each matrix is positive definite. Thus $A+B$ is positive definite, and its eigenvalues are all positive. The proof does not require A and B to commute or share eigenvectors; adding their eigenvalue lists would not be a valid general argument.

Exercise 7.2.28: The inverse of a positive definite matrix

An orthogonal diagonalization $A = PDP^T$ has positive diagonal entries λ_i. Its inverse is $A^{-1} = PD^{-1}P^T$, with positive eigenvalues $1/\lambda_i$. Hence its quadratic form is positive definite. For a direct check, given $x \neq 0$ let $y = A^{-1}x \neq 0$. Since A is symmetric, $x^TA^{-1}x = (Ay)^Ty = y^TAy > 0$. Thus both the eigenvalue argument and the defining inequality give the claim.

7.3 Constrained optimization

A quadratic form is an expression $Q(x) = x^T Ax$, where we take A to be symmetric. A coefficient of $x_i x_j$ with $i \neq j$ must be divided equally between a_{ij} and a_{ji}. For example, $6x_1x_2$ means $a_{12} = a_{21} = 3$, not 6.

The central argument. The spectral theorem gives an orthonormal eigenbasis $u_1, \ldots, u_n$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. Put $P = [u_1 \ \cdots \ u_n]$ and $x = Py$. Then $P^TP = I$, so

$$x^Tx = y^Ty, \qquad Q(x) = \sum_{j=1}^{n} \lambda_j y_j^2.$$

On the unit sphere, the nonnegative numbers y_j^2 add to one. Thus

$$\lambda_n \leq Q(x) \leq \lambda_1.$$

Taking $x = u_1$ attains the upper bound. If we also require $x^Tu_1 = 0$, then $y_1 = 0$, and the maximum becomes λ_2, attained at u_2. Eigenvalues here are listed *with multiplicities*; a repeated largest eigenvalue can remain the largest value after one direction is excluded. These observations explain the optimization steps below without requiring multivariable differentiation.

Practice problem 7.3.P1: Remove the mixed term using an orthonormal basis

What we must show. Diagonalize $3x_1^2 + 3x_2^2 + 2x_1x_2$ by $x = Py$. The symmetric matrix is $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Its characteristic polynomial is $(3-\lambda)^2 - 1$, with roots $4, 2$. For 4, solving $(A - 4I)v = 0$ gives $v_2 = v_1$; choose $(1,1)^T$. For 2, the equations give $v_2 = -v_1$; choose $(-1,1)^T$. Each vector has length $\sqrt{2}$, so use

$$P = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

Its columns have length one and dot product zero. Therefore $P^TAP = \operatorname{diag}(4, 2)$, and $Q(Py) = 4y_1^2 + 2y_2^2$.

Conclusion. The displayed P gives the requested substitution. Normalizing the eigenvectors is essential to making P orthogonal.

Practice problem 7.3.P2: Translate the maximizing direction back to the original variables

For a unit x, the preceding change of variables gives $y_1^2 + y_2^2 = 1$. Consequently $Q(x) = 4y_1^2 + 2y_2^2 \leq 4$. Equality holds for $y = (1, 0)^T$. But the requested vector is x, not y: multiplying by P gives $x = (1, 1)^T/\sqrt{2}$. Its squared length is $(1 + 1)/2 = 1$.

Conclusion. The maximum is 4, attained at $(1, 1)^T/\sqrt{2}$. Its negative also attains the same value.

Exercise 7.3.1: An orthogonal substitution in three variables

What we must show. Obtain the diagonal coefficients $9, 6, 3$ for the given quadratic form.

Build the matrix and find its eigenvectors.

The coefficients give

$$A = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 6 & -2 \\ 0 & -2 & 7 \end{bmatrix}.$$

Expanding $\det(\lambda I - A)$ gives $(\lambda - 9)(\lambda - 6)(\lambda - 3)$. Solving the corresponding homogeneous systems gives the convenient eigenvectors

$$v_9 = (-1, -2, 2)^T, \quad v_6 = (2, 1, 2)^T, \quad v_3 = (-2, 2, 1)^T.$$

For example, at $\lambda = 9$ the first equation is $-4a + 2b = 0$, so $b = 2a$; the last is $-2b - 2c = 0$, so $c = -b$. Taking $a = -1$ produces v_9. The other two vectors are checked by multiplying by A. Each has squared length 9, and their pairwise dot products are zero.

Normalize and put the columns in the requested order.

Set

$$P = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Then $P^T P = I$ and $AP = P\,\mathrm{diag}(9, 6, 3)$. Multiplying this last identity on the left by P^T gives $P^T AP = \mathrm{diag}(9, 6, 3)$.

Conclusion. With $x = Py$, the form is $9y_1^2 + 6y_2^2 + 3y_3^2$.

Exercise 7.3.2: Include the missing zero coefficient

What we must show. Obtain the stated diagonal form, keeping three coordinates. The symmetric matrix is

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}, \qquad \det(\lambda I - A) = \lambda(\lambda - 2)(\lambda - 5).$$

Corresponding eigenvectors are $(1, 1, 1)^T$, $(-2, 1, 1)^T$, and $(0, -1, 1)^T$, for $5, 2, 0$, respectively. Their squared lengths are $3, 6, 2$, and their pairwise dot products vanish. Thus take

$$P = \begin{bmatrix} 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix}.$$

Direct multiplication gives $P^T AP = \mathrm{diag}(5, 2, 0)$.

Conclusion. $x = Py$ transforms the form to $5y_1^2 + 2y_2^2 + 0y_3^2$. The zero coefficient does not remove a coordinate from the substitution.

Exercise 7.3.3: Exclude one principal direction

(a) Bound the form on all unit vectors.

Exercise 7.3.1 gives $Q(x) = 9y_1^2 + 6y_2^2 + 3y_3^2$ and $y_1^2 + y_2^2 + y_3^2 = 1$. Replacing 6 and 3 by 9 shows $Q(x) \le 9$.

(b) Give a maximizing unit vector in the original coordinates.

Take $u = (-1, -2, 2)^T/3$, the first column of P. Then $Au = 9u$ and $u^T u = 1$, so $Q(u) = u^T(9u) = 9$.

(c) Impose the orthogonality condition.

Because $x = Py$ and $u = Pe_1$, we have $x^T u = y^T e_1 = y_1$. Thus $x^T u = 0$ forces $y_1 = 0$. Now $Q(x) = 6y_2^2 + 3y_3^2 \le 6(y_2^2 + y_3^2) = 6$. Equality holds at the second column $(2, 1, 2)^T/3$ of P.

Conclusion. The answers are 9, $u = (-1, -2, 2)^T/3$, and 6.

Exercise 7.3.4: The same constraints with a null direction

Use the orthogonal coordinates of Exercise 7.3.2. On the unit sphere, $Q(x) = 5y_1^2 + 2y_2^2 \le 5(y_1^2 + y_2^2 + y_3^2) = 5$. The unit vector $u = (1, 1, 1)^T/\sqrt{3}$ has $Au = 5u$, so it attains 5. The additional equation $x^T u = 0$ is $y_1 = 0$. Therefore $Q(x) = 2y_2^2 \le 2$, with equality at $(-2, 1, 1)^T/\sqrt{6}$.

Conclusion. (a) 5; (b) $u = (1, 1, 1)^T/\sqrt{3}$; (c) 2.

Exercise 7.3.5: A negative mixed coefficient

The matrix is $A = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$. The equation $(5 - \lambda)^2 - 4 = 0$ gives eigenvalues $7, 3$. Their unit eigenvectors are $u = (-1, 1)^T/\sqrt{2}$ and $v = (1, 1)^T/\sqrt{2}$. They are perpendicular. Writing $x = au + bv$ gives $\|x\|^2 = a^2 + b^2$ and $Q(x) = 7a^2 + 3b^2$. Hence the unrestricted unit maximum is 7, attained at u. When $x^T u = 0$, we have $a = 0$ and $b^2 = 1$, so the value is 3.

Conclusion. (a) 7; (b) $(-1, 1)^T/\sqrt{2}$; (c) 3.

Exercise 7.3.6: Remember to halve an odd mixed coefficient

Here $A = \begin{bmatrix} 7 & 3/2 \\ 3/2 & 3 \end{bmatrix}$, since the mixed term is $3x_1x_2$. Its characteristic equation is

$$(7 - \lambda)(3 - \lambda) - \frac{9}{4} = 0,$$

whose roots are $15/2$ and $5/2$. At $15/2$, the first row of $(A - \lambda I)v = 0$ gives $-v_1 + 3v_2 = 0$; choose $(3, 1)^T$. Its unit version is $u = (3, 1)^T/\sqrt{10}$. A perpendicular unit eigenvector is $v = (-1, 3)^T/\sqrt{10}$, for $5/2$. For $x = au + bv$, the unit constraint gives $a^2 + b^2 = 1$, and $Q(x) = (15/2)a^2 + (5/2)b^2$. Requiring $x^T u = 0$ sets $a = 0$.

Conclusion. (a) $15/2$; (b) $(3, 1)^T/\sqrt{10}$; (c) $5/2$.

Exercise 7.3.7: Normalize the largest-eigenvalue vector

The given coefficients give

$$A = \begin{bmatrix} -2 & 2 & 0 \\ 2 & -1 & 2 \\ 0 & 2 & 0 \end{bmatrix}.$$

Its eigenvalues are $2, -1, -4$, so the maximum on the unit sphere is 2. To find a maximizing vector, solve $(A - 2I)v = 0$. The first row gives $-4v_1 + 2v_2 = 0$, hence $v_2 = 2v_1$. The last gives $2v_2 - 2v_3 = 0$, hence $v_3 = v_2$. Taking $v_1 = 1$ gives $v = (1, 2, 2)^T$, with length 3. Consequently $x = v/3$ is unit, and $Q(x) = x^T(2x) = 2$.

Conclusion. One required vector is $(1, 2, 2)^T/3$.

Exercise 7.3.8: A repeated largest eigenvalue

The matrix is

$$A = \begin{bmatrix} 7 & -4 & -2 \\ -4 & 1 & -4 \\ -2 & -4 & 7 \end{bmatrix}.$$

Its eigenvalues are $9, 9, -3$. At 9, every row equation reduces to $v_1 + 2v_2 + v_3 = 0$. Choose $v = (-1, 0, 1)^T$ in this plane and normalize: $x = (-1, 0, 1)^T/\sqrt{2}$. The squared length is one, $Ax = 9x$, and $Q(x) = 9$. No unit vector can exceed the largest eigenvalue.

Conclusion. $(-1, 0, 1)^T/\sqrt{2}$ is a maximizing unit vector; the maximum is 9.

Exercise 7.3.9: Find the maximum without finding a vector

The mixed coefficient -2 gives $A = \begin{bmatrix} 7 & -1 \\ -1 & 3 \end{bmatrix}$. Its characteristic equation is

$$(7 - \lambda)(3 - \lambda) - 1 = \lambda^2 - 10\lambda + 20 = 0.$$

The quadratic formula gives $\lambda = 5 \pm \sqrt{5}$. By the unit-sphere bound established at the start, the maximum is the larger root. No eigenvector is needed for the question asked.

Conclusion. The maximum is $5 + \sqrt{5}$.

Exercise 7.3.10: A form with both positive and negative values

Now $A = \begin{bmatrix} -3 & -1 \\ -1 & 5 \end{bmatrix}$. The equation is

$$(-3 - \lambda)(5 - \lambda) - 1 = \lambda^2 - 2\lambda - 16 = 0,$$

with roots $1 \pm \sqrt{17}$. The constraint is exactly the unit-sphere constraint, so the larger eigenvalue gives the maximum.

Conclusion. The maximum is $1 + \sqrt{17}$.

Exercise 7.3.11: Evaluate a form at a unit eigenvector

If $Ax = 3x$ and $x^T x = 1$, substitute the eigenvector equation first:

$$x^T Ax = x^T(3x) = 3(x^T x) = 3.$$

This particular calculation does not require symmetry of A; the two stated equations are sufficient.

Conclusion. The value is 3.

Exercise 7.3.12: Every eigenvalue lies between the extreme unit values

Let λ be an eigenvalue of the real symmetric matrix A. The spectral theorem supplies a nonzero real eigenvector v. Normalize it: $x = v/\|v\|$. Then $x^T x = 1$ and $Ax = \lambda x$, so $x^T Ax = \lambda x^T x = \lambda$. By definition, m is a lower bound and M is an upper bound for the values of the form at all unit vectors. They therefore bound this particular value.

Conclusion. $m \le \lambda \le M$. Normalization places the eigenvector on the very set over which the extrema are defined.

Exercise 7.3.13: Every intermediate value is attained

What we must show. Given $m \le t \le M$, construct a unit x with $x^T Ax = t$.

First separate the constant case.

If $M = m$, then $t = m$ and any unit eigenvector for m works. The proposed expression can use $\alpha = 0$, giving $x = u_n$. This also handles the one-dimensional case without demanding two perpendicular vectors in a one-dimensional space.

For $M > m$, choose the interpolation weight explicitly.

Put $\alpha = (t - m)/(M - m)$. The inequalities on t imply $0 \le \alpha \le 1$, and rearrangement gives $t = (1 - \alpha)m + \alpha M$. Let $Au_1 = Mu_1$ and $Au_n = mu_n$, where the vectors are unit. Distinct eigenvalues of a symmetric matrix have orthogonal eigenvectors: indeed $u_1^T Au_n = mu_1^T u_n$ and, by symmetry, the same expression equals $(Au_1)^T u_n = Mu_1^T u_n$. Since $M - m \neq 0$, $u_1^T u_n = 0$.

Check both the constraint and the desired value.

Define $x = \sqrt{1 - \alpha}\, u_n + \sqrt{\alpha}\, u_1$. Expanding gives

$$x^T x = (1 - \alpha)\underbrace{u_n^T u_n}_{1} + \alpha \underbrace{u_1^T u_1}_{1} + 2\sqrt{\alpha(1 - \alpha)}\underbrace{u_n^T u_1}_{0} = 1.$$

The mixed terms in $x^T Ax$ also vanish, because, for example, $u_n^T Au_1 = Mu_n^T u_1 = 0$. The remaining terms give

$$x^T Ax = (1 - \alpha)m + \alpha M = t.$$

Conclusion. Every number in $[m, M]$ occurs. The square roots in x make the squared coefficients, rather than the coefficients, add to one.

Exercise 7.3.14: An exact computer-assisted four-dimensional example

Convert the mixed terms to a symmetric matrix.

$$A = \frac{1}{2}\begin{bmatrix} 0 & 1 & 3 & 30 \\ 1 & 0 & 30 & 3 \\ 3 & 30 & 0 & 1 \\ 30 & 3 & 1 & 0 \end{bmatrix}.$$

To check its spectrum without trusting rounded output, use the four pairwise perpendicular vectors

$$(1,1,1,1)^T, \quad (1,-1,-1,1)^T, \quad (-1,1,-1,1)^T, \quad (-1,-1,1,1)^T.$$

Multiplication by A gives these same vectors times $17, 13, -14, -16$, respectively. Each has length 2; dividing them by 2 gives an orthonormal eigenbasis and verifies the entire spectrum.

Answer the three optimization questions.

The maximum on unit vectors is 17, attained at $u = (1,1,1,1)^T/2$. Orthogonality to u removes its coordinate, leaving the next largest eigenvalue, 13. It is attained at $(1,-1,-1,1)^T/2$.

Conclusion. (a) 17; (b) $(1,1,1,1)^T/2$; (c) 13.

Exercise 7.3.15: A negative constrained maximum

$$A = \frac{1}{2}\begin{bmatrix} 0 & 3 & 5 & 7 \\ 3 & 0 & 7 & 5 \\ 5 & 7 & 0 & 3 \\ 7 & 5 & 3 & 0 \end{bmatrix}.$$

The same four orthogonal sign vectors from Exercise 7.3.14 are eigenvectors, now with eigenvalues $15/2, -1/2, -5/2, -9/2$, in that order. For instance, the first row sum is $(3+5+7)/2 = 15/2$, verifying the first coordinate of the all-ones eigenvector equation; the other rows have the same sum. Checking the other three sign vectors verifies the remaining eigenvalues. Their unit versions form a basis, so no eigenvalue is omitted. Thus the unit-sphere maximum is $15/2$, at $u = (1,1,1,1)^T/2$. On $u^\perp$, the largest remaining eigenvalue is $-1/2$, attained at $(1,-1,-1,1)^T/2$. A maximum need not be positive.

Conclusion. (a) $15/2$; (b) $(1,1,1,1)^T/2$; (c) $-1/2$.

Exercise 7.3.16: One diagonal term and six mixed terms

The symmetric matrix and a factored characteristic polynomial are

$$A = \begin{bmatrix} 4 & -3 & -5 & -5 \\ -3 & 0 & -3 & -3 \\ -5 & -3 & 0 & -1 \\ -5 & -3 & -1 & 0 \end{bmatrix},$$

$$\det(\lambda I - A) = (\lambda - 9)(\lambda - 3)(\lambda - 1)(\lambda + 9).$$

For an exact check, vectors $(-2,0,1,1)^T$, $(1,-3,1,1)^T$, $(0,0,-1,1)^T$, and $(1,1,1,1)^T$ have eigenvalues $9, 3, 1, -9$, respectively. Each assertion follows by one matrix–vector multiplication, and the vectors

are nonzero and pairwise perpendicular. The first has length $\sqrt{6}$, so choose $u = (-2, 0, 1, 1)^T/\sqrt{6}$. Then $Q(u) = 9$. With $x \perp u$, the maximum becomes 3, attained at $(1, -3, 1, 1)^T/\sqrt{12}$.

Conclusion. (a) 9; (b) $(-2, 0, 1, 1)^T/\sqrt{6}$; (c) 3.

Exercise 7.3.17: The largest eigenvalue can itself be negative

Here

$$A = \begin{bmatrix} -6 & -2 & -2 & -2 \\ -2 & -10 & 0 & 0 \\ -2 & 0 & -13 & 3 \\ -2 & 0 & 3 & -13 \end{bmatrix}.$$

Its characteristic polynomial is $(\lambda + 4)(\lambda + 10)(\lambda + 12)(\lambda + 16)$. Nonzero eigenvectors for $-4, -10, -12, -16$, respectively, are

$$(-3, 1, 1, 1)^T, \quad (0, -2, 1, 1)^T, \quad (1, 1, 1, 1)^T, \quad (0, 0, -1, 1)^T.$$

They are pairwise perpendicular. Checking $Av = \lambda v$ for each one verifies the spectrum independently of an eigenvalue routine. The largest eigenvalue is -4, since $-4 > -10 > -12 > -16$; one does *not* choose the largest absolute value. The first vector has squared length 12. After excluding its unit direction, -10 is the largest remaining eigenvalue, attained at $(0, -2, 1, 1)^T/\sqrt{6}$.

Conclusion. (a) -4; (b) $(-3, 1, 1, 1)^T/\sqrt{12}$; (c) -10.

Reproducing the computer checks in 14–17. Enter each displayed matrix using exact fractions, not rounded decimals. For a symmetric matrix, the following Python/SymPy commands return an exact spectrum and check a chosen vector. Replace `entries`, `vector`, and `eigenvalue` with the data displayed in the corresponding solution:

```
from sympy import Matrix, Rational, eye
A = Matrix(entries)
print(A.eigenvals())
v = Matrix(vector)
lam = eigenvalue
assert A == A.T
assert A*v == lam*v
u = v / (v.dot(v))**Rational(1, 2)
assert (u.T*u)[0] == 1
```

The exact eigenvector certificates above also give a complete verification without installing software. The matrices, results, and certificates are all specified; the computer problems do not require an external dataset.

7.4 The singular value decomposition

For an $m \times n$ real matrix, a full singular value decomposition (SVD) is $A = U\Sigma V^T$, where U and V are square orthogonal matrices and Σ is rectangular diagonal with nonnegative entries in decreasing order. The singular values are the square roots of the eigenvalues of A^TA. Positive singular values count rank. The distinction between V and the displayed factor V^T is essential when reading singular vectors.

Right singular vectors satisfy $A^TAv_i = \sigma_i^2 v_i$; when $\sigma_i > 0$, set $u_i = Av_i/\sigma_i$. Complete the left columns orthonormally if needed. This method is useful for understanding the theory; for ill-conditioned large matrices a direct numerical SVD avoids squaring the condition number by forming A^TA.

Practice problem 7.4.P1: Transpose an SVD and track zero conventions

Transposing the factorization gives

$$A^T = (U\Sigma V^T)^T = V\Sigma^T U^T.$$

Both outer matrices are orthogonal, and Σ^T is rectangular diagonal with the same nonnegative diagonal entries. Thus it is an SVD of the transpose. The positive singular values of A and A^T coincide, including multiplicities. If singular values are listed as all square roots of the eigenvalues of A^TA, the number of additional zero entries can differ for a rectangular matrix because A^TA and AA^T have different sizes. There is no disagreement about the nonzero singular values.

Exercise 7.4.1: Nonnegative singular values

Compute the positive-semidefinite matrix

$$A^TA = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}.$$

Its eigenvalues are $9,\ 1$. Singular values are their nonnegative square roots, listed in decreasing order:

$$(3,\ 1).$$

A negative diagonal entry of A does not make a negative singular value: lengths and square roots here are nonnegative.

Exercise 7.4.2: Nonnegative singular values

Compute the positive-semidefinite matrix

$$A^TA = \begin{bmatrix} 25 & 0 \\ 0 & 0 \end{bmatrix}.$$

Its eigenvalues are $25,\ 0$. Singular values are their nonnegative square roots, listed in decreasing order:

$$(5,\ 0).$$

A negative diagonal entry of A does not make a negative singular value: lengths and square roots here are nonnegative.

Exercise 7.4.3: Nonnegative singular values

Compute the positive-semidefinite matrix

$$A^TA = \begin{bmatrix} 6 & \sqrt{6} \\ \sqrt{6} & 7 \end{bmatrix}.$$

Its eigenvalues are $9,\ 4$. Singular values are their nonnegative square roots, listed in decreasing order:

$$(3,\ 2).$$

A negative diagonal entry of A does not make a negative singular value: lengths and square roots here are nonnegative.

Exercise 7.4.4: Nonnegative singular values

Compute the positive-semidefinite matrix

$$A^TA = \begin{bmatrix} 3 & 2\sqrt{3} \\ 2\sqrt{3} & 7 \end{bmatrix}.$$

Its eigenvalues are $9,\ 1$. Singular values are their nonnegative square roots, listed in decreasing order:

$$(3,\ 1).$$

A negative diagonal entry of A does not make a negative singular value: lengths and square roots here are nonnegative.

Exercise 7.4.5: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} -3 & 0 \\ 0 & 0 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^TA = \begin{bmatrix} 9 & 0 \\ 0 & 0 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $9,\ 0$. Their nonnegative square roots are $3,\ 0$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^TV = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 3$ gives

$$u_1 = \frac{Av_1}{3} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^2$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The dimensions are 2×2, 2×2, and 2×2, respectively. Exact multiplication gives $U^TU = I$, $V^TV = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.6: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^TA = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $4,\ 1$. Their nonnegative square roots are $2,\ 1$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^TV = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 2$ gives

$$u_1 = \frac{Av_1}{2} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

For $i = 2$, $\sigma_i = 1$ gives

$$u_2 = \frac{Av_2}{1} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The dimensions are 2×2, 2×2, and 2×2, respectively. Exact multiplication gives $U^TU = I$, $V^TV = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.7: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^TA = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $9,\ 4$. Their nonnegative square roots are $3,\ 2$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} \frac{2\sqrt{5}}{5} & -\frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^TV = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 3$ gives

$$u_1 = \frac{Av_1}{3} = \begin{bmatrix} \frac{\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = 2$ gives

$$u_2 = \frac{Av_2}{2} = \begin{bmatrix} -\frac{2\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} \end{bmatrix}.$$

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{\sqrt{5}}{5} & -\frac{2\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}, \qquad V^T = \begin{bmatrix} \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \\ -\frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

The dimensions are 2×2, 2×2, and 2×2, respectively. Exact multiplication gives $U^TU = I$, $V^TV = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.8: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^TA = \begin{bmatrix} 4 & 6 \\ 6 & 13 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $16,\ 1$. Their nonnegative square roots are $4,\ 1$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} \frac{\sqrt{5}}{5} & -\frac{2\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^TV = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 4$ gives

$$u_1 = \frac{Av_1}{4} = \begin{bmatrix} \frac{2\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = 1$ gives

$$u_2 = \frac{Av_2}{1} = \begin{bmatrix} -\frac{\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{2\sqrt{5}}{5} & -\frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \qquad V^T = \begin{bmatrix} \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \\ -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \end{bmatrix}.$$

The dimensions are 2×2, 2×2, and 2×2, respectively. Exact multiplication gives $U^TU = I$, $V^TV = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.9: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 7 & 1 \\ 0 & 0 \\ 5 & 5 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 74 & 32 \\ 32 & 26 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $90,\ 10$. Their nonnegative square roots are $3\sqrt{10},\ \sqrt{10}$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} \frac{2\sqrt{5}}{5} & -\frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 3\sqrt{10}$ gives

$$u_1 = \frac{Av_1}{3\sqrt{10}} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = \sqrt{10}$ gives

$$u_2 = \frac{Av_2}{\sqrt{10}} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^3$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 3\sqrt{10} & 0 \\ 0 & \sqrt{10} \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \\ -\frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

The dimensions are 3×3, 3×2, and 2×2, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.10: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 4 & -2 \\ 2 & -1 \\ 0 & 0 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 20 & -10 \\ -10 & 5 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $25,\ 0$. Their nonnegative square roots are $5,\ 0$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 5$ gives

$$u_1 = \frac{Av_1}{5} = \begin{bmatrix} -\frac{2\sqrt{5}}{5} \\ -\frac{\sqrt{5}}{5} \\ 0 \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^3$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{5} & -\frac{2\sqrt{5}}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 5 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

The dimensions are 3×3, 3×2, and 2×2, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.11: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $90,\ 0$. Their nonnegative square roots are $3\sqrt{10},\ 0$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} -\frac{3\sqrt{10}}{10} & \frac{\sqrt{10}}{10} \\ \frac{\sqrt{10}}{10} & \frac{3\sqrt{10}}{10} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 3\sqrt{10}$ gives

$$u_1 = \frac{Av_1}{3\sqrt{10}} = \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^3$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{1}{3} & \frac{2\sqrt{2}}{3} & 0 \\ -\frac{2}{3} & \frac{\sqrt{2}}{6} & \frac{\sqrt{2}}{2} \\ -\frac{2}{3} & \frac{\sqrt{2}}{6} & -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} -\frac{3\sqrt{10}}{10} & \frac{\sqrt{10}}{10} \\ \frac{\sqrt{10}}{10} & \frac{3\sqrt{10}}{10} \end{bmatrix}.$$

The dimensions are 3×3, 3×2, and 2×2, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.12: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $3,\ 2$. Their nonnegative square roots are $\sqrt{3},\ \sqrt{2}$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = \sqrt{3}$ gives

$$u_1 = \frac{Av_1}{\sqrt{3}} = \begin{bmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = \sqrt{2}$ gives

$$u_2 = \frac{Av_2}{\sqrt{2}} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^3$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{3}}{3} & 0 & -\frac{\sqrt{6}}{3} \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The dimensions are 3×3, 3×2, and 2×2, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.13: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $25,\ 9,\ 0$. Their nonnegative square roots are $5,\ 3,\ 0$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{6} & -\frac{2}{3} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{6} & \frac{2}{3} \\ 0 & \frac{2\sqrt{2}}{3} & \frac{1}{3} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 5$ gives

$$u_1 = \frac{Av_1}{5} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = 3$ gives

$$u_2 = \frac{Av_2}{3} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{6} & -\frac{\sqrt{2}}{6} & \frac{2\sqrt{2}}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}.$$

The dimensions are 2×2, 2×3, and 3×3, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.14: A unit vector attaining the largest stretch

For any unit vector x, expand it in the orthonormal right singular basis: $x = c_1v_1 + c_2v_2$, where $c_1^2 + c_2^2 = 1$. Exercise 7 gives singular values 3 and 2, so

$$\|Ax\|^2 = 9c_1^2 + 4c_2^2 \le 9(c_1^2 + c_2^2) = 9.$$

The largest possible length is therefore 3, attained at the first right singular vector

$$x = \begin{bmatrix} \frac{2\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} \end{bmatrix}.$$

It is a unit vector, and direct multiplication gives $\|Ax\| = 3$. Its negative is another maximizer.

Exercise 7.4.15: Read rank and subspace bases without reconstruction

(a) Count positive singular values.

The displayed diagonal factor has positive entries 7.10 and 3.10 and then zero, so $\operatorname{rank} A = 2$.

(b) Read the appropriate columns of the two orthogonal factors.

A basis of $\operatorname{Col} A$ is given by the first two columns of U:

$$\left\{ \begin{bmatrix} .40 \\ .37 \\ -.84 \end{bmatrix}, \begin{bmatrix} -.78 \\ -.33 \\ -.52 \end{bmatrix} \right\}.$$

For the null space take column 3 of V, which is *row 3 of the printed V^T factor*:

$$\operatorname{Nul} A = \operatorname{Span} \left\{ \begin{bmatrix} .58 \\ -.58 \\ .58 \end{bmatrix} \right\}.$$

The entries are rounded descriptions of the true singular vectors; they need not give exact orthogonality or exact zero residuals when interpreted as exact terminating decimals. The requested structural information comes from the stated SVD, not from recomputing rank after rounding.

Exercise 7.4.16: A rectangular SVD with a two-dimensional null space

The diagonal factor has two positive entries, 12.48 and 6.34, so $\operatorname{rank} A = 2$. Since A has four columns, its nullity is $4 - 2 = 2$. A basis of $\operatorname{Col} A$ is supplied by the first two U columns:

$$\left\{ \begin{bmatrix} -.86 \\ .31 \\ .41 \end{bmatrix}, \begin{bmatrix} -.11 \\ .68 \\ -.73 \end{bmatrix} \right\}.$$

The last two columns of V are the last two rows of the printed right-hand factor V^T. Thus a basis of $\operatorname{Nul} A$ is

$$\left\{ \begin{bmatrix} .65 \\ .08 \\ -.16 \\ -.73 \end{bmatrix}, \begin{bmatrix} -.34 \\ .42 \\ -.84 \\ -.08 \end{bmatrix} \right\}.$$

As in Exercise 15, the printed decimal columns are approximations to the exact basis vectors associated with the given SVD.

Exercise 7.4.17: SVD consequences and proofs

Invertibility makes every singular value positive. Reverse the factor order to obtain

$$A^{-1} = V\Sigma^{-1}U^T.$$

This has orthogonal outer factors and positive diagonal entries $1/\sigma_1, \ldots, 1/\sigma_n$. If an SVD is required to list singular values in decreasing order, reverse this order: let J be the permutation matrix reversing the standard basis. Then

$$A^{-1} = (VJ)(J^T\Sigma^{-1}J)(UJ)^T.$$

Its diagonal is $1/\sigma_n \geq \cdots \geq 1/\sigma_1$, as required. In particular the largest singular value of the inverse is the reciprocal of the smallest singular value of A.

Exercise 7.4.18: SVD consequences and proofs

Taking determinants of $A = U\Sigma V^T$ gives $\det A = (\det U)(\det \Sigma)(\det V^T)$. Orthogonal matrices have determinants ± 1, while $\det \Sigma = \sigma_1 \cdots \sigma_n \geq 0$. Taking absolute values proves $|\det A| = \prod_{j=1}^{n} \sigma_j$. If A is singular, a singular value is zero and both sides are zero.

Exercise 7.4.19: SVD consequences and proofs

Use orthogonality in the products:

$$A^TA = V\Sigma^TU^TU\Sigma V^T = V(\Sigma^T\Sigma)V^T,$$

$$AA^T = U\Sigma V^TV\Sigma^TU^T = U(\Sigma\Sigma^T)U^T.$$

The middle matrices are diagonal. Their positive entries are $\sigma_1^2, \ldots, \sigma_r^2$, with zeros in the remaining positions. Thus the columns of V and U are the respective eigenvectors, with those squared eigenvalues; their nonnegative square roots are the singular values. For a rectangular matrix, the numbers of structural zeros on the two sides can differ.

Exercise 7.4.20: SVD consequences and proofs

A real positive definite matrix in this section is symmetric. Its orthogonal diagonalization $A = PDP^T$ has strictly positive eigenvalues on the diagonal of D. Thus the factors already satisfy the SVD requirements, with $U = V = P$ and $\Sigma = D$. If necessary reorder the eigenvectors and diagonal entries together so that the positive diagonal is decreasing. Positive definiteness is what avoids any sign adjustments to make singular values nonnegative.

Exercise 7.4.21: SVD consequences and proofs

Because $P^TP = I_m$,

$$(PA)^T(PA) = A^TP^TPA = A^TA.$$

The two Gram matrices are identical, not merely similar. Hence their eigenvalues and their nonnegative square roots agree, proving the singular values agree. Geometrically an orthogonal transformation of the output does not change any length.

Exercise 7.4.22: SVD consequences and proofs

Assume there is a second right singular direction, so $n \geq 2$. For a unit $x \perp v_1$, expand $x = \sum_{j=2}^{n} c_j v_j$ with $\sum_{j=2}^{n} c_j^2 = 1$. Orthogonality of the left singular vectors gives

$$\|Ax\|^2 = \sum_{j=2}^{n} \sigma_j^2 c_j^2 \leq \sigma_2^2 \sum_{j=2}^{n} c_j^2 = \sigma_2^2.$$

The bound is attained at $x = v_2$. Thus the maximum length is σ_2, including the cases of repeated singular values or $\sigma_2 = 0$. The order $\sigma_1 \geq \sigma_2 \geq \cdots$ is used in the inequality.

Exercise 7.4.23: SVD consequences and proofs

The product $U\Sigma$ has first r columns $\sigma_j u_j$ and all other columns zero. Row j of V^T is v_j^T. By the column-times-row expansion of a matrix product,

$$U\Sigma V^T = \sum_{j=1}^{r} (\sigma_j u_j) v_j^T = \sum_{j=1}^{r} \sigma_j u_j v_j^T.$$

Every term is an $m \times n$ rank-one matrix. The dimensions match, and zero-singular-value terms contribute nothing.

Exercise 7.4.24: SVD consequences and proofs

Transpose the SVD and use $U^T u_j = e_j$:

$$A^T u_j = V\Sigma^T U^T u_j = V\Sigma^T e_j = V(\sigma_j e_j) = \sigma_j v_j \qquad (1 \leq j \leq r).$$

Thus multiplication by A^T sends a positive left singular direction to its matching right singular direction, with the same scale.

Exercise 7.4.25: SVD consequences and proofs

Find the standard matrix A of T and compute a full SVD $A = U\Sigma V^T$. Let $\mathcal{B}$ be the ordered columns of V in the domain, and let $\mathcal{C}$ be the ordered columns of U in the codomain. They are bases because the matrices are orthogonal. The coordinate matrix is

$$[T]_{\mathcal{C} \leftarrow \mathcal{B}} = U^{-1} A V = U^T A V = \Sigma.$$

This is the requested rectangular diagonal matrix, including zero rows or columns where dimensions and rank require them. Using different bases on input and output is what permits this for every rectangular linear map.

Exercise 7.4.26: Numerical SVD with measured reconstruction checks

What we must show. Compute an SVD of

$$A = \begin{bmatrix} -18 & 13 & -4 & 4 \\ 2 & 19 & -4 & 12 \\ -14 & 11 & -12 & 8 \\ -2 & 21 & 4 & 8 \end{bmatrix}.$$

Following the construction in the examples, compute eigenvectors of A^TA in decreasing eigenvalue order. Take the positive square roots to obtain the singular values, then set $u_i = Av_i/\sigma_i$. Exact row reduction confirms rank 3, so exactly that many singular values are positive. The full-precision calculation gives factors whose entries, rounded to two decimals as requested, are

$$U \approx \begin{bmatrix} 0.50 & -0.50 & -0.50 & -0.50 \\ 0.50 & 0.50 & 0.50 & -0.50 \\ 0.50 & -0.50 & 0.50 & 0.50 \\ 0.50 & 0.50 & -0.50 & 0.50 \end{bmatrix}.$$

$$\Sigma \approx \begin{bmatrix} 40.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 20.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 10.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 0.00 \end{bmatrix}.$$

$$V^T \approx \begin{bmatrix} -0.40 & 0.80 & -0.20 & 0.40 \\ 0.80 & 0.40 & 0.40 & 0.20 \\ 0.40 & -0.20 & -0.80 & 0.40 \\ -0.20 & -0.40 & 0.40 & 0.80 \end{bmatrix}.$$

Before rounding, the Frobenius reconstruction error $\|U\Sigma V^T - A\|_F$ was 2.35×10^{-14}, and the orthogonality errors for U, V were 2.48×10^{-15} and 8.2×10^{-16}. Rounding the displayed entries increases residuals and need not retain exact orthogonality; the decomposition is specified numerically to two decimals, not as an exact factorization of the rounded factors. A reproducible implementation of the stated construction is:

```
import numpy as np
lam,V = np.linalg.eigh(A.T@A)
idx = np.argsort(lam)[::-1]
lam,V = lam[idx],V[:,idx]
r = 3 if A.shape[1] == 4 else 4  # exact ranks here
s = np.sqrt(lam[:r])
U = (A@V[:,:r])/s
if r < A.shape[0]:
    Q,_ = np.linalg.qr(U,mode='complete')
    U = np.column_stack((U,Q[:,r:]))
Sigma = np.zeros(A.shape)
Sigma[np.arange(r),np.arange(r)] = s
print(U,Sigma,V.T)
```

Use floating-point arrays for the displayed input. Eigenvector signs may differ between libraries; changing the matching left and right signs does not change the product. A direct numerical SVD was also used to check the computed singular values.

Exercise 7.4.27: Numerical SVD with measured reconstruction checks

What we must show. Compute an SVD of

$$A = \begin{bmatrix} 6 & -8 & -4 & 5 & -4 \\ 2 & 7 & -5 & -6 & 4 \\ 0 & -1 & -8 & 2 & 2 \\ -1 & -2 & 4 & 4 & -8 \end{bmatrix}.$$

Following the construction in the examples, compute eigenvectors of A^TA in decreasing eigenvalue order. Take the positive square roots to obtain the singular values, then set $u_i = Av_i/\sigma_i$. Exact row reduction confirms rank 4, so exactly that many singular values are positive. The full-precision calculation gives factors whose entries, rounded to two decimals as requested, are

$$U \approx \begin{bmatrix} -0.57 & 0.65 & 0.42 & 0.27 \\ 0.63 & 0.24 & 0.68 & -0.29 \\ 0.07 & 0.63 & -0.53 & -0.56 \\ -0.51 & -0.34 & 0.29 & -0.73 \end{bmatrix}.$$

$$\Sigma \approx \begin{bmatrix} 16.46 & 0.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 12.16 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 4.87 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 4.31 & 0.00 \end{bmatrix}.$$

$$V^T \approx \begin{bmatrix} -0.10 & 0.61 & -0.21 & -0.52 & 0.55 \\ 0.39 & -0.29 & -0.84 & 0.14 & 0.19 \\ 0.74 & 0.27 & 0.07 & -0.38 & -0.49 \\ 0.41 & -0.50 & 0.45 & -0.23 & 0.58 \\ -0.36 & -0.48 & -0.19 & -0.72 & -0.29 \end{bmatrix}.$$

Before rounding, the Frobenius reconstruction error $\|U\Sigma V^T - A\|_F$ was 1.13×10^{-14}, and the orthogonality errors for U, V were 2.99×10^{-15} and 1.04×10^{-15}. Rounding the displayed entries increases residuals and need not retain exact orthogonality; the decomposition is specified numerically to two decimals, not as an exact factorization of the rounded factors. A reproducible implementation of the stated construction is:

```
import numpy as np
lam,V = np.linalg.eigh(A.T@A)
idx = np.argsort(lam)[::-1]
lam,V = lam[idx],V[:,idx]
r = 3 if A.shape[1] == 4 else 4  # exact ranks here
s = np.sqrt(lam[:r])
U = (A@V[:,:r])/s
if r < A.shape[0]:
    Q,_ = np.linalg.qr(U,mode='complete')
    U = np.column_stack((U,Q[:,r:]))
Sigma = np.zeros(A.shape)
Sigma[np.arange(r),np.arange(r)] = s
print(U,Sigma,V.T)
```

Use floating-point arrays for the displayed input. Eigenvector signs may differ between libraries; changing the matching left and right signs does not change the product. A direct numerical SVD was also used to check the computed singular values.

Exercise 7.4.28: Singular values and a condition number

The matrix referred to from Section 2.3 is

$$A = \begin{bmatrix} 4 & 0 & -3 & -7 \\ -6 & 9 & 9 & 9 \\ 7 & -5 & 10 & 19 \\ -1 & 2 & 4 & -1 \end{bmatrix}.$$

A stable direct SVD, rather than forming the potentially ill-conditioned Gram matrix numerically, gives the singular values

$$(25.96092922,\ 14.45655782,\ 4.669971173,\ 2.286801047).$$

All are positive (also $\det A = -4008 \neq 0$ in exact arithmetic). The spectral condition number is therefore

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_4} \approx 11.3525089.$$

The computation is reproducible with

```
s = np.linalg.svd(A,compute_uv=False)
print(s,s[0]/s[-1])
```

The ratio measures the greatest stretching relative to the least stretching. A large ratio explains sensitivity to perturbations even though the matrix is exactly invertible.

Exercise 7.4.29: Singular values and a condition number

The matrix referred to from Section 2.3 is

$$A = \begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 6 & 4 & 2 & 8 & -8 \\ 7 & 5 & 3 & 10 & 9 \\ 9 & 6 & 4 & -9 & -5 \\ 8 & 5 & 2 & 11 & 4 \end{bmatrix}.$$

A stable direct SVD, rather than forming the potentially ill-conditioned Gram matrix numerically, gives the singular values

$$(25.9343222,\ 16.7554367,\ 11.29172511,\ 1.078525439,\ 0.0003779276369).$$

All are positive (also $\det A = 2 \neq 0$ in exact arithmetic). The spectral condition number is therefore

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_5} \approx 68622.45484.$$

The computation is reproducible with

```
s = np.linalg.svd(A,compute_uv=False)
print(s,s[0]/s[-1])
```

The ratio measures the greatest stretching relative to the least stretching. A large ratio explains sensitivity to perturbations even though the matrix is exactly invertible.

7.5 Applications to image processing and statistics

A data matrix has one observation in each *column* and one measured variable in each row. Its mean column is $M = N^{-1}\sum_{k=1}^{N} X_k$. Subtract M from every column to form B; then the sample covariance matrix is $S = BB^T/(N-1)$, for $N > 1$.

For a unit coefficient vector c, the scalar index $y = c^T X$ has variance $c^T Sc$. The greatest possible variance is the largest eigenvalue of S, attained at a corresponding unit eigenvector. Principal component scores normally use the centered observation $X - M$. Centering changes an index by a constant and hence does not change its variance. Signs of principal component directions are arbitrary: reversing a unit eigenvector reverses every score but preserves its explained variance.

Practice problem 7.5.P1: Center the weight-height data and calculate covariance

The five-column mean is $M = (130, 65)^T$. Subtracting it from every observation gives

$$B = \begin{bmatrix} -10 & -5 & -5 & 5 & 15 \\ -4 & -5 & -1 & 3 & 7 \end{bmatrix}.$$

Each row sums to zero, checking the centering step. The cross-product is

$$BB^T = \begin{bmatrix} 400 & 190 \\ 190 & 100 \end{bmatrix}.$$

Since the sample has $N = 5$, divide by $N - 1 = 4$:

$$S = \frac{1}{4}BB^T = \begin{bmatrix} 100 & \frac{95}{2} \\ \frac{95}{2} & 25 \end{bmatrix}.$$

Its diagonal entries are the weight and height sample variances; its off-diagonal entry is their sample covariance.

Practice problem 7.5.P2: A one-number size index

Diagonalizing the covariance matrix gives variances approximately 123.01859218 and 1.98140782. A unit direction for the larger one is

$$c \approx \begin{bmatrix} 0.899901 \\ 0.436094 \end{bmatrix}.$$

Thus a centered size score is $y \approx 0.899901(w - 130) + 0.436094(h - 65)$. Its variance is the larger eigenvalue. The total variance is $\operatorname{tr} S = 125$, so the fraction retained is 0.98414874, or about 98.4149%. This index uses the original units specified by the question; standardizing the two variables to unit variance would define a different PCA problem.

Exercise 7.5.1: Center all observations and form the sample covariance

Compute the mean of each row using all six observations.

$$M = \begin{bmatrix} 12 \\ 10 \end{bmatrix}.$$

Subtract that column from every observation.

$$B = \begin{bmatrix} 7 & 10 & -6 & -9 & -10 & 8 \\ 2 & -4 & -1 & 5 & 3 & -5 \end{bmatrix}.$$

Each row sums to zero. This checks that no observation was omitted and that the rows and columns were not reversed.

Form the sample covariance with denominator $6 - 1$.

$$BB^T = \begin{bmatrix} 430 & -135 \\ -135 & 80 \end{bmatrix}, \qquad S = \frac{1}{5} BB^T = \begin{bmatrix} 86 & -27 \\ -27 & 16 \end{bmatrix}.$$

For example, the off-diagonal entry is the sum of six products of paired centered measurements, divided by 5. The two diagonal entries are sums of squares divided by 5, so they are nonnegative as variances must be.

Exercise 7.5.2: Center all observations and form the sample covariance

Compute the mean of each row using all six observations.

$$M = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

Subtract that column from every observation.

$$B = \begin{bmatrix} -3 & 1 & -2 & 2 & 3 & -1 \\ -6 & 2 & -3 & -1 & 6 & 2 \end{bmatrix}.$$

Each row sums to zero. This checks that no observation was omitted and that the rows and columns were not reversed.

Form the sample covariance with denominator $6 - 1$.

$$BB^T = \begin{bmatrix} 28 & 40 \\ 40 & 90 \end{bmatrix}, \qquad S = \frac{1}{5} BB^T = \begin{bmatrix} \frac{28}{5} & 8 \\ 8 & 18 \end{bmatrix}.$$

For example, the off-diagonal entry is the sum of six products of paired centered measurements, divided by 5. The two diagonal entries are sums of squares divided by 5, so they are nonnegative as variances must be.

Exercise 7.5.3: Find both principal component directions

Use the exact covariance matrix from Exercise 1:

$$S = \begin{bmatrix} 86 & -27 \\ -27 & 16 \end{bmatrix}.$$

Its characteristic equation is $t^2 - 102t + 647 = 0$. In decreasing order its eigenvalues are approximately

$$\lambda_1 = 95.20407221, \qquad \lambda_2 = 6.79592779.$$

Choose the following orthonormal eigenvector columns, in that same order:

$$P \approx \begin{bmatrix} 0.946515 & 0.322659 \\ -0.322659 & 0.946515 \end{bmatrix}.$$

The two principal component scores are the entries of $Y = P^T(X - M)$, with mean

$$M = \begin{bmatrix} 12 \\ 10 \end{bmatrix}.$$

Their covariance matrix is diagonal, $P^T S P = \operatorname{diag}(\lambda_1, \lambda_2)$, so they are uncorrelated. The first score explains about 93.33733% of total variance; the second explains the remainder. A sign-reversed column gives an equivalent principal component.

Exercise 7.5.4: Find both principal component directions

Use the exact covariance matrix from Exercise 2:

$$S = \begin{bmatrix} \frac{28}{5} & 8 \\ 8 & 18 \end{bmatrix}.$$

Its characteristic equation is $\frac{5t^2 - 118t + 184}{5} = 0$. In decreasing order its eigenvalues are approximately

$$\lambda_1 = 21.92126474, \qquad \lambda_2 = 1.67873526.$$

Choose the following orthonormal eigenvector columns, in that same order:

$$P \approx \begin{bmatrix} 0.440130 & 0.897934 \\ 0.897934 & -0.440130 \end{bmatrix}.$$

The two principal component scores are the entries of $Y = P^T(X - M)$, with mean

$$M = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

Their covariance matrix is diagonal, $P^T S P = \operatorname{diag}(\lambda_1, \lambda_2)$, so they are uncorrelated. The first score explains about 92.88672% of total variance; the second explains the remainder. A sign-reversed column gives an equivalent principal component.

Exercise 7.5.5: A leading principal component from a supplied covariance matrix

The full covariance matrix needed for this calculation is supplied in the exercise:

$$S = \begin{bmatrix} \frac{4103}{25} & \frac{3273}{100} & \frac{2026}{25} \\ \frac{3273}{100} & \frac{13486}{25} & \frac{24913}{100} \\ \frac{2026}{25} & \frac{24913}{100} & \frac{18911}{100} \end{bmatrix}.$$

No original satellite image is required to diagonalize this matrix. Compute its symmetric eigenproblem and normalize the eigenvector of largest eigenvalue. The eigenvalues, in decreasing order, are approximately

$$(677.49712437,\ 181.58112062,\ 33.59175501).$$

A maximizing unit direction is

$$c \approx \begin{bmatrix} 0.129554 \\ 0.874423 \\ 0.467547 \end{bmatrix}.$$

Thus $y_1 = c_1\widehat{x}_1 + c_2\widehat{x}_2 + c_3\widehat{x}_3$, where the hats denote mean-deviation variables, is the first principal component. The uncentered combination has the same variance. The maximum variance is approximately 677.49712437, and total variance is $\operatorname{tr} S = \frac{89267}{100}$. Therefore the explained fraction is

$$\frac{\lambda_1}{\operatorname{tr} S} \approx 0.75895586, \quad \text{or } 75.89559\%.$$

The actual computation used the real symmetric eigenvalue routine:

```
eigenvalues,P = np.linalg.eigh(S)
c = P[:,-1]
print(c,eigenvalues[-1],
      100*eigenvalues[-1]/np.trace(S))
```

The maximum eigenvalue is the last one because `eigh` returns increasing order. A reversed eigenvector sign is equivalent. Orthogonality and the residual $Sc - \lambda_1 c$ were checked at full precision before rounding.

Exercise 7.5.6: A leading principal component from a supplied covariance matrix

The full covariance matrix needed for this calculation is supplied in the exercise:

$$S = \begin{bmatrix} \frac{741}{25} & \frac{919}{50} & 5 \\ \frac{919}{50} & \frac{1041}{50} & \frac{703}{50} \\ 5 & \frac{703}{50} & \frac{2921}{100} \end{bmatrix}.$$

No original satellite image is required to diagonalize this matrix. Compute its symmetric eigenproblem and normalize the eigenvector of largest eigenvalue. The eigenvalues, in decreasing order, are approximately

$$(51.69566979,\ 24.53510921,\ 3.43922100).$$

A maximizing unit direction is

$$c \approx \begin{bmatrix} 0.615525 \\ 0.599424 \\ 0.511683 \end{bmatrix}.$$

Thus $y_1 = c_1\widehat{x}_1 + c_2\widehat{x}_2 + c_3\widehat{x}_3$, where the hats denote mean-deviation variables, is the first principal component. The uncentered combination has the same variance. The maximum variance is approximately 51.69566979, and total variance is $\operatorname{tr} S = \frac{7967}{100}$. Therefore the explained fraction is

$$\frac{\lambda_1}{\operatorname{tr} S} \approx 0.64887247, \quad \text{or } 64.88725\%.$$

The actual computation used the real symmetric eigenvalue routine:

```
eigenvalues,P = np.linalg.eigh(S)
c = P[:,-1]
print(c,eigenvalues[-1],
      100*eigenvalues[-1]/np.trace(S))
```

The maximum eigenvalue is the last one because `eigh` returns increasing order. A reversed eigenvector sign is equivalent. Orthogonality and the residual $Sc - \lambda_1 c$ were checked at full precision before rounding.

Exercise 7.5.7: A maximal-variance scalar combination

For the data in Exercise 1, the covariance matrix is

$$S = \begin{bmatrix} 86 & -27 \\ -27 & 16 \end{bmatrix}.$$

Choose the unit coefficient direction

$$c \approx \begin{bmatrix} 0.946515 \\ -0.322659 \end{bmatrix}.$$

The requested uncentered variable is $y_1 \approx (0.946515)x_1 + (-0.322659)x_2$. Its variance is $c^T S c = \lambda_1 \approx 95.20407221$, the maximum by the quadratic-form theorem. Using $x_1 - M_1, x_2 - M_2$ instead would simply subtract its mean and leave that variance unchanged. The total variance is $\operatorname{tr} S = 102$, so the explained percentage is approximately 93.33733%. Both the unit-length constraint and the covariance eigenvector equation were checked before rounding the coefficients.

Exercise 7.5.8: A maximal-variance scalar combination

For the data in Exercise 2, the covariance matrix is

$$S = \begin{bmatrix} \frac{28}{5} & 8 \\ 8 & 18 \end{bmatrix}.$$

Choose the unit coefficient direction

$$c \approx \begin{bmatrix} 0.440130 \\ 0.897934 \end{bmatrix}.$$

The requested uncentered variable is $y_1 \approx (0.440130)x_1 + (0.897934)x_2$. Its variance is $c^T S c = \lambda_1 \approx 21.92126474$, the maximum by the quadratic-form theorem. Using $x_1 - M_1, x_2 - M_2$ instead would simply subtract its mean and leave that variance unchanged. The total variance is $\operatorname{tr} S = \frac{118}{5}$, so the explained percentage is approximately 92.88672%. Both the unit-length constraint and the covariance eigenvector equation were checked before rounding the coefficients.

Exercise 7.5.9: A unit eigenvector maximizes the index variance

The sample variance of $y = c^T X$ is $c^T S c$, with $c^T c = 1$. The characteristic roots of the supplied symmetric matrix are 9, 6, 3, with multiplicity. Its largest eigenvalue is 9. Solving $(S - (9)I)c = 0$ and normalizing gives

$$c = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}.$$

The exact checks are $c^T c = 1$ and $Sc = (9)c$. Hence the index

$$y = (\frac{1}{3})x_1 + (\frac{2}{3})x_2 + (\frac{2}{3})x_3$$

has the largest possible variance, 9. This accounts for the fraction $\frac{1}{2}$ of the total variance 18.

Exercise 7.5.10: A unit eigenvector maximizes the index variance

The sample variance of $y = c^T X$ is $c^T S c$, with $c^T c = 1$. The characteristic roots of the supplied symmetric matrix are 15, 3, 3, with multiplicity. Its largest eigenvalue is 15. Solving $(S - (15)I)c = 0$ and normalizing gives

$$c = \begin{bmatrix} \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \end{bmatrix}.$$

The exact checks are $c^T c = 1$ and $Sc = (15)c$. Hence the index

$$y = (\frac{\sqrt{6}}{6})x_1 + (\frac{\sqrt{6}}{3})x_2 + (\frac{\sqrt{6}}{6})x_3$$

has the largest possible variance, 15. This accounts for the fraction $\frac{5}{7}$ of the total variance 21.

Exercise 7.5.11: Covariance identities and their dimensions

(a) The transformed mean remains zero.

Write $B = [X_1 \ \cdots \ X_N]$ and $C = [Y_1 \ \cdots \ Y_N] = P^T B$. Since the original mean is zero, $\sum_k X_k = 0$. Thus

$$\frac{1}{N} \sum_k Y_k = P^T \left(\frac{1}{N} \sum_k X_k \right) = 0.$$

Equivalently, $B\mathbf{1} = 0$ implies $C\mathbf{1} = P^T B\mathbf{1} = 0$. No orthogonality of P is required.

(b) Compute the new covariance using part (a).

Because both matrices are already centered,

$$S_Y = \frac{1}{N-1} CC^T = \frac{1}{N-1} P^T BB^T P = P^T SP.$$

The transpose on C^T reverses factors and gives $C^T = B^T P$, which explains the rightmost P.

Exercise 7.5.12: Covariance identities and their dimensions

For orthogonal P, the relation $X = PY$ is equivalent to $Y = P^T X$. Centering commutes with that linear change of variables, and Exercise 11 gives covariance $S_Y = P^T SP$. Total variance is the trace. The cyclic trace identity yields

$$\operatorname{tr}(S_Y) = \operatorname{tr}(P^T SP) = \operatorname{tr}(SPP^T) = \operatorname{tr}(SI) = \operatorname{tr}(S).$$

Thus orthogonal changes redistribute variance among new coordinate directions without creating or destroying its total. This proof also covers data not originally centered, because covariance first subtracts the transformed mean.

Exercise 7.5.13: Covariance identities and their dimensions

Let $M = N^{-1} \sum_{k=1}^{N} X_k$ and $B = [X_1 - M \ \cdots \ X_N - M]$. Multiply the column partition of B by the row partition of B^T:

$$BB^T = \sum_{k=1}^{N} (X_k - M)(X_k - M)^T.$$

Hence the sample covariance is

$$\boxed{S = \frac{1}{N-1} \sum_{k=1}^{N} (X_k - M)(X_k - M)^T.}$$

Every summand is a $p \times p$ outer-product matrix. Its (i, j) entry is the product of the centered measurements of variables i and j in observation k. When $p = 1$, the outer product is just the square $(t_k - m)^2$, giving exactly the usual scalar sample-variance formula. The sum has N terms, one per observation; the lowercase n printed as an upper bound in the displayed scalar formula should be read as that sample size.

7.S Chapter 7 supplementary exercises

These exercises connect symmetric spectral decompositions, quadratic-form positivity, Gram matrices, singular value decompositions, and minimum-length solutions. A symmetric matrix is positive definite when $x^TAx > 0$ for every $x \neq 0$, not merely for one vector. A pseudoinverse provides a least-squares solution even when an exact solution does not exist; that distinction is important in the final two printed exercises.

Exercise 7.S.1: Seventeen true–false statements

(a) **True.** If $A = PDP^T$ with P orthogonal and D diagonal, then $A^T = PD^TP^T = A$.

(b) **False.** The 90° rotation $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is orthogonal but is not symmetric.

(c) **True.** $\|Ax\|^2 = x^TA^TAx = x^Tx$ for orthogonal A. Taking nonnegative square roots gives the length equality.

(d) **False.** Principal axes require mutually perpendicular eigenvector directions. For $A = I_2$, every invertible P diagonalizes A, but $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has nonorthogonal columns, so its columns do not give orthogonal principal axes.

(e) **False.** Orthogonal columns need not be unit columns. For $P = 2I$, $P^T = 2I \neq I/2 = P^{-1}$.

(f) **False.** All coefficients of $x_1^2 + x_2^2 + 4x_1x_2$ are positive, but its value at $(1, -1)$ is -2.

(g) **False.** $x_1^2 - x_2^2$ is positive at e_1 and negative at e_2, so positivity at one input does not give definiteness.

(h) **True.** The symmetric coefficient matrix admits an orthonormal eigenbasis, and its orthogonal change of variables removes all mixed terms by the Principal Axes Theorem.

(i) **False.** For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, the largest diagonal entry is 1, but the unit vector $(1, 1)^T/\sqrt{2}$ gives $x^TAx = 2$, the largest eigenvalue.

(j) **False without a constraint.** A positive definite form is unbounded above: $Q(tx) = t^2Q(x) \to \infty$ for fixed $x \neq 0$. The greatest eigenvalue is the maximum only when $\|x\| = 1$ is imposed.

(k) **False.** If P is orthogonal, nonzero u gives $x = Pu \neq 0$, so $(Pu)^TA(Pu) > 0$. An invertible variable change preserves positive definiteness rather than turning it negative.

(l) **False.** An indefinite form has a symmetric coefficient matrix with both positive and negative eigenvalues. “Not definite eigenvalues” is not a mathematical alternative definition; semidefinite forms are not indefinite merely because an eigenvalue is zero.

(m) **True.** Substitution gives $(Pu)^TA(Pu) = u^TP^TAPu = u^T(P^{-1}AP)u$.

(n) **False.** Orthonormal columns are required for UU^T to be the projection formula. For $U = 2e_1$ and $x = e_1$, $UU^Tx = 4e_1$, whereas the projection onto its column line is e_1.

(o) **True.** An SVD gives $\|Bx\|^2 = \sum_i \sigma_i^2 y_i^2 \le \sigma_1^2 \sum_i y_i^2 = \sigma_1^2$ for $y = V^T x$ and $\|x\| = 1$.

(p) **True as an existence statement.** From an ordinary SVD $B = U\Sigma V^T$, take $P = U$ and $Q = V^T$. The transpose of an orthogonal matrix is orthogonal, so $B = P\Sigma Q$ has exactly the asserted form. This does not mean one can omit a transpose while keeping the same right singular-vector matrix V; the new symbol Q denotes its transpose.

(q) **False.** The singular values of $A^T A$ are the squares of those of A. For $A = 2I$, they are 4 versus 2, so the lists differ.

Exercise 7.S.2: Build a symmetric matrix with prescribed eigenvectors

(a) Each outer product satisfies $(u_i u_i^T)^T = u_i u_i^T$. A real linear combination of symmetric matrices is symmetric, so $A^T = A$.

(b) For each j, orthonormality gives

$$Au_j = \sum_i \lambda_i u_i (u_i^T u_j) = \lambda_j u_j.$$

Every u_j is nonzero, so it is an eigenvector with eigenvalue λ_j. Because these n vectors are a basis, the list contains all eigenvalues, including their multiplicities.

Exercise 7.S.3: Rank counts the nonzero spectral summands

Write $A = PDP^T$ with orthogonal P. Multiplication by invertible matrices preserves rank, so $\operatorname{rank} A = \operatorname{rank} D$, the number of nonzero diagonal eigenvalues. Thus exactly r terms in $A = \sum_i \lambda_i u_i u_i^T$ are nonzero. For each such term, every column is a multiple of u_i, and at least one column is nonzero because $u_i \neq 0$ and $\lambda_i \neq 0$. Each nonzero term therefore has rank one. Discarding zero terms represents A as a sum of r rank-one matrices, as claimed. When $r = 0$, this is the empty sum representing zero.

Exercise 7.S.4: An orthogonal column/null decomposition for a symmetric matrix

(a) A vector z is perpendicular to every column of A precisely when $A^T z = 0$. Since $A^T = A$, this says $z \in \operatorname{Nul} A$. Therefore $(\operatorname{Col} A)^\perp = \operatorname{Nul} A$.

(b) Orthogonally decompose y relative to $W = \operatorname{Col} A$: $y = \widehat{y} + z$, where $\widehat{y} \in \operatorname{Col} A$ and $z \in (\operatorname{Col} A)^\perp = \operatorname{Nul} A$. Such a decomposition exists by the Orthogonal Decomposition Theorem and is unique.

Exercise 7.S.5: A nonzero-eigenvalue eigenvector lies in the column space

If $Av = \lambda v$ with $\lambda \neq 0$, then $v = A(v/\lambda)$. This explicitly writes v as an output of A, so $v \in \operatorname{Col} A$. The nonzero-eigenvalue condition permits division; a zero-eigenvalue eigenvector need not belong to the column space.

Exercise 7.S.6: Construct the decomposition using an eigenbasis

Choose an orthonormal eigenbasis u_i and write $y = \sum_i c_i u_i$. Separate the sum into

$$\widehat{y} = \sum_{\lambda_i \neq 0} c_i u_i, \qquad z = \sum_{\lambda_i = 0} c_i u_i.$$

Every summand of $\widehat{y}$ belongs to $\operatorname{Col} A$ by Exercise 5, so $\widehat{y} \in \operatorname{Col} A$. Every summand of z is annihilated by A, so $z \in \operatorname{Nul} A$. Their sum is y, giving the requested second proof. The two sums are perpendicular because they use disjoint parts of an orthonormal basis.

Exercise 7.S.7: Prove the Cholesky characterization

A Cholesky factorization implies positive definiteness.

If $A = R^T R$ with invertible real R, it is symmetric and $x^T A x = \|Rx\|^2 > 0$ whenever $x \neq 0$. Thus A is positive definite.

A positive definite matrix supplies such a factorization.

Exercise 7.2.26 gives an invertible positive definite square root B with $A = B^T B$. Take a QR factorization $B = QR$, choosing the signs of columns of Q and rows of R so the nonzero diagonal entries of R are positive. Here Q is square orthogonal and R is upper triangular. Then $A = B^T B = R^T Q^T Q R = R^T R$. The full rank of B makes R invertible, and the sign choices give the required positive diagonal. This proves both directions of the equivalence.

Exercise 7.S.8: Convert Cholesky into an LU factorization

From Exercise 7, $A = R^T R$ with upper triangular R and positive r_{ii}. Let $D = \operatorname{diag}(r_{11}, \ldots, r_{nn})$. Define $L = R^T D^{-1}$, which is lower triangular with diagonal entries one, and $U = DR$, which is upper triangular with diagonal entries $r_{ii}^2 > 0$. Then $LU = R^T D^{-1} DR = R^T R = A$, as required. The parenthetical converse in the text is understood for symmetric matrices and the usual unit-lower-triangular LU convention. Without symmetry, an arbitrary upper triangular matrix with positive diagonal need not be a positive definite *symmetric* matrix.

Exercise 7.S.9: A Gram matrix has the same rank as its original matrix

$G = A^T A$ is symmetric and $x^T G x = \|Ax\|^2 \geq 0$, so it is positive semidefinite. Moreover, $Gx = 0$ implies $0 = x^T G x = \|Ax\|^2$, hence $Ax = 0$. Conversely $Ax = 0$ immediately implies $Gx = A^T 0 = 0$. Thus $\operatorname{Nul} G = \operatorname{Nul} A$. Both matrices have the same number n of columns, so rank–nullity gives $\operatorname{rank} G = n - \dim \operatorname{Nul} G = n - \dim \operatorname{Nul} A = \operatorname{rank} A$. This also proves why forming a Gram matrix does not create new independent column directions in exact arithmetic.

Exercise 7.S.10: Construct a rank-revealing Gram factor

A symmetric positive semidefinite G has r positive eigenvalues $\lambda_1, \ldots, \lambda_r$ and the rest zero. Let U_r contain corresponding orthonormal eigenvectors, and $D_r = \operatorname{diag}(\lambda_1, \ldots, \lambda_r)$. Then $G = U_r D_r U_r^T$.

Choose the $r \times n$ matrix

$$\boxed{A = D_r^{1/2} U_r^T}.$$

Its Gram matrix is $A^T A = U_r D_r U_r^T = G$. Equivalently the hinted factor $B = U_r D_r^{1/2}$ satisfies $G = BB^T$ and $A = B^T$. When $r = 0$, $G = 0$ and the $0 \times n$ empty matrix has zero Gram matrix under the empty-sum convention.

Exercise 7.S.11: A polar decomposition, including singular matrices

Take a square SVD $A = U\Sigma V^T$, with U, V orthogonal and Σ diagonal with nonnegative entries. Set

$$P = U\Sigma U^T, \qquad Q = UV^T.$$

Then P is symmetric positive semidefinite by its orthogonal diagonalization, and $\operatorname{rank} P = \operatorname{rank} \Sigma = \operatorname{rank} A$. Also $Q^T Q = VU^T UV^T = I$, so Q is orthogonal. Finally $PQ = U\Sigma U^T UV^T = U\Sigma V^T = A$. No inverse singular values were used, so the construction works for singular A as well. The orthogonal factor need not be unique in that case.

Exercise 7.S.12: The pseudoinverse produces two orthogonal projections

Use the reduced SVD $A = U_r D V_r^T$ and $A^+ = V_r D^{-1} U_r^T$, where the r singular values in D are positive.

(a) Since $V_r^T V_r = I_r$, $AA^+ = U_r D D^{-1} U_r^T = U_r U_r^T$. The columns of U_r are an orthonormal basis of $\operatorname{Col} A$; therefore $AA^+ y$ is the orthogonal projection of y onto that space.

(b) Similarly $A^+ A = V_r V_r^T$, the projection onto $\operatorname{Col} V_r = \operatorname{Row} A$.

(c) Multiplying the formulas and using orthonormality gives

$$AA^+ A = U_r U_r^T U_r D V_r^T = A, \qquad A^+ AA^+ = V_r V_r^T V_r D^{-1} U_r^T = A^+.$$

Thus both requested triple-product identities hold.

Exercise 7.S.13: A consistent system's unique minimum-length solution

(a) Since $b = Ax$ for some x by consistency, $x^+ = A^+ b = A^+ Ax$ is the projection of x onto $\operatorname{Row} A$ by Exercise 12(b). Hence $x^+ \in \operatorname{Row} A$.

(b) Since $b \in \operatorname{Col} A$, its projection there is itself, so $Ax^+ = AA^+ b = b$ by Exercise 12(a). Thus x^+ is indeed a solution.

(c) For any other solution u, $A(u - x^+) = b - b = 0$. Thus $u - x^+ \in \operatorname{Nul} A$, perpendicular to $x^+ \in \operatorname{Row} A = (\operatorname{Nul} A)^\perp$. Pythagoras yields $\|u\|^2 = \|x^+\|^2 + \|u - x^+\|^2 \geq \|x^+\|^2$. Equality holds only when $u = x^+$. Taking square roots proves the minimum length and its uniqueness.

Exercise 7.S.14: The minimum-length least-squares solution for arbitrary data

Let $\widehat{b} = AA^+ b$, the projection of b onto $\operatorname{Col} A$. A vector x is a least-squares solution of $Ax = b$ exactly when $Ax = \widehat{b}$, since $\widehat{b}$ is the unique closest output. This new equation is consistent. By Exercise

13 its unique minimum-length solution is $A^+\widehat{b}$. The triple-product identity from Exercise 12 gives $A^+\widehat{b} = A^+AA^+b = A^+b$. Thus $\boxed{A^+b}$ is the unique minimum-length least-squares solution, even when the original equation itself has no solution.

Exercise 7.S.15: A pseudoinverse and an inconsistent printed right side

Check consistency before claiming to solve the original system.

Rows 3 and 4 of the printed A are identical, but the prescribed $b = (6, -1, -4, 6)^T$ has unequal entries 3 and 4. Hence $Ax = b$ has *no exact solution*. The requested pseudoinverse calculation is nevertheless well defined: it returns the minimum-length least-squares solution from Exercise 14, not an exact solution from Exercise 13. We retain the printed data and make that distinction explicit.

Compute from an SVD, then verify the exact rational matrix.

An executed numerical SVD has nonzero singular values approximately 9.84443449, 2.62465501, and 1.09466692. Its fourth singular value is at roundoff level, representing an exact zero. In NumPy the construction can be reproduced by

```
U,d,Vt = np.linalg.svd(A,full_matrices=False)
r = np.count_nonzero(d > 1e-12)
Ap = (Vt[:r,:].T/d[:r])@U[:,:r].T
```

The exact rational result, verified by the four pseudoinverse identities, is

$$A^+ = \frac{1}{40}\begin{bmatrix} -2 & -14 & 13 & 13 \\ -2 & -14 & 13 & 13 \\ -2 & 6 & -7 & -7 \\ 2 & -6 & 7 & 7 \\ 4 & -12 & -6 & -6 \end{bmatrix}.$$

Here the verification is $AA^+A = A$, $A^+AA^+ = A^+$, and symmetry of AA^+ and A^+A; all products were checked with exact rational arithmetic. Multiplying by the printed right side gives

$$\widehat{x} = A^+b = (7/10, 7/10, -4/5, 4/5, 3/5)^T, \qquad A\widehat{x} = (6, -1, 1, 1)^T.$$

The residual is $(0, 0, -5, 5)^T$ and is perpendicular to every column of A, since its two nonzero components multiply identical rows with opposite weights. Thus $A^T(b - A\widehat{x}) = 0$, independently checking the least-squares claim.

Check row-space membership and the strict norm inequality.

An explicit witness is $\widehat{x} = A^T(7/100, -91/100, 129/100, 0)^T$, so $\widehat{x} \in \operatorname{Row} A$. Choose $u = (-1, 1, 0, 0, 0)^T$. Its first two columns cancel, so $Au = 0$; also $\widehat{x} \cdot u = 0$. Finally $\|\widehat{x}\|^2 = 131/50$ and $\|\widehat{x} + u\|^2 = 231/50$, proving $\|\widehat{x}\| < \|\widehat{x} + u\|$.

Exercise 7.S.16: A second minimum-length least-squares computation

Again distinguish least squares from exact consistency.

Here row 4 is three times row 3, whereas the prescribed right side has $6 \neq 3(-4)$. Thus the printed system is also inconsistent. Its pseudoinverse solution must be interpreted via Exercise 14.

Construct and verify the pseudoinverse.

The executed SVD's nonzero singular values are approximately $12.9536237, 1.44552724, .337763285$. Using the SVD formula in Exercise 15 and verifying exact products gives

$$A^+ = \frac{1}{20}\begin{bmatrix} 10 & 0 & -1 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 40 & 10 & 30 \\ 10 & -20 & -7 & -21 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

All four pseudoinverse identities hold exactly. The resulting input and fitted output are

$$\widehat{x} = (23/10, 0, 5, -9/10, 0)^T, \qquad A\widehat{x} = (6, -1, 7/5, 21/5)^T.$$

The residual is $(0, 0, -27/5, 9/5)^T$. Its row-weight combination vanishes because row 4 equals three times row 3: $(-27/5)\,\mathrm{row}_3\, A + (9/5)\,\mathrm{row}_4\, A = 0$. Hence the normal equations hold and this is a least-squares solution.

Verify the two geometric checks.

The identity $\widehat{x} = A^T(7/10, 109/10, 27, 0)^T$ places it in the row space. Take the nonzero null vector $u = e_2 = (0, 1, 0, 0, 0)^T$, since column 2 is zero. It is perpendicular to $\widehat{x}$, whose second coordinate is zero. The squared norms are $\|\widehat{x}\|^2 = 311/10$ and $\|\widehat{x} + u\|^2 = 321/10$, giving the requested strict inequality. These checks establish the minimum-length property without falsely claiming that the inconsistent original equation has been solved exactly.